



SR 816.111

Ergänzung 2.1 zu Anhang 5 der Verordnung des EDI vom 22. März 2017 über das elektronische Patientendossier

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## Nationale Integrationsprofile nach Artikel 5 Absatz 1 Buchstabe c EPDV-EDI

### Authorization Decision Request (CH:ADR) and Privacy Policy Query (CH:PPQ)

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Ergänzung 2.1 zu Anhang 5 EPDV-EDI : Nationale Anpassungen

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# 1 Introduction

Dieses Dokument wurde erstellt, um die schweizerischen Regelungen der Verordnung über das elektronische Patientendossier (EPDV, SR 816.11) zu erfüllen. Die EPDV und die EPDV-EDI (SR 816.111) werden in der Amtlichen Sammlung veröffentlicht (in Deutsch, Französisch und Italienisch)<sup>1</sup>.

Das elektronische Patientendossier (EPD) ist ein System mit mehreren (Stamm-)Gemeinschaften. Die Patientin oder der Patient stimmt nicht nur der Eröffnung und Nutzung des EPD zu, sie oder er kann über das Patientenportal auch explizit Zugriffsregeln festlegen sowie bestimmen, ob Gesundheitsfachpersonen ihnen erteilte Berechtigungen an andere Gesundheitsfachpersonen delegieren dürfen. Diese Zugriffsregeln werden in der Stammgemeinschaft der Patientin oder des Patienten gespeichert.

Die IHE-Akteure der (Stamm-)Gemeinschaften, die diese Zugriffsrechte durchsetzen müssen (z.B. Dokumentenregister), müssen deshalb als Policy Enforcing Service Provider tätig sein und einen XACML Policy Enforcement Point (PEP) implementieren. Diese Akteure müssen Schnittstellen implementieren, um Zugriffsentscheidungen vom XACML Policy Decision Point (PDP) der Stammgemeinschaft der Patientin oder des Patienten abzufragen.

Die Komplexität und Flexibilität der definierten Zugriffsregeln, welche Patientinnen oder Patienten gesetzlich wahrnehmen können, erfordern, dass die Zugangsportale für Patientinnen und Patienten sowie Zugangsportale für Gesundheitsfachpersonen als CH:PPQ oder CH:PPQm Policy Source und Consumer fungieren, um die Zugriffsregeln im CH:PPQ oder CH:PPQm Policy Repository, einem XACML Policy Administration Point (PAP), hinzuzufügen, abzufragen, zu aktualisieren und zu löschen. Die Interoperabilitätsstandards für die Verwaltung und Durchsetzung der Zugriffsrechte für Patientinnen und Patienten sind in diesem Dokument beschrieben.

The Swiss Electronic Patient Record (EPR) is a system with multiple (reference) communities, where the patient not only consents to the creation and usage of the EPR. The patient can also explicitly define access rules through the patient portal and specify whether healthcare professionals are allowed to delegate obtained permissions to other healthcare professionals. These access rules are stored in the patient's reference community.

IHE Actors of the (reference) communities having to enforce these access policies (e.g., Document Registries) shall therefore act as a Policy Enforcing Service Provider implementing an XACML Policy Enforcement Point (PEP). These actors may need to implement interfaces to request policy decisions from the XACML Policy Decision Point (PDP) of the patient's reference community.

The complexity and flexibility of access rule definitions that were granted to patients by law, require the patient portals and healthcare professional portals to act as CH:PPQ or CH:PPQm Policy Source and Consumer to add, query, update and delete policies in the CH:PPQ or CH:PPQm Policy Repository, an XACML Policy Administration Point (PAP). The interoperability standards to manage the patient's access rules and to enforce them, are specified in this document.

This document fulfills the Swiss regulations of the Ordinance on the Electronic Patient Record (EPRO, SR 816.11). The EPRO and the EPRO-FDHA (SR 816.111) are published in Official Compilation of Federal Legislation (available in German, French and Italian)<sup>1</sup>.

<sup>1</sup> German: <https://www.admin.ch/opc/de/classified-compilation/20111795/index.html>;  
French: <https://www.admin.ch/opc/fr/classified-compilation/20111795/index.html>;  
Italian: <https://www.admin.ch/opc/it/classified-compilation/20111795/index.html>.

## 2 Volume 1 – Integration Profiles

### 2.1 Authorization Decision Request [CH:ADR]

According to Swiss EPR regulations, patients have the right to decide who is allowed to access and modify data in their EPR and under which circumstance (cf. emergency access).

The access restrictions concern following actors:

- a. XDS Document Registry;
- b. XDS Document Repository;
- c. MHD Document Recipient;
- d. MHD Document Responder;
- e. RMU Restricted Metadata Update Responder;
- f. CH:PPQ Policy Repository;
- g. CH:PPQm Policy Repository;
- h. CH:ATC Patient Audit Record Repository;
- i. XDS-I.b Imaging Document Source;

Each of them is grouped with a XUA X-Service Provider or IUA Resource Server, which means that they shall be able to

- j. use the CH:XUA assertion or CH:IUA token obtained from the XUA X-Service User or IUA Authorization Server to determine whether the user who initiated the transaction is allowed to access the requested information stored in a patient's EPR;
- k. and interpret (enforce) this decision.

Exactly this is the purpose of the Authorization Decision Request (CH:ADR) profile: it computes authorization decisions and defines how to enforce them for each particular request type.

The CH:ADR profile is an adaptation of the IHE Secure Retrieve (SeR) profile meeting the EPR requirements and using the same technology stack: SAML 2.0 profile of XACML v2.0 as data format and Web Services according to IHE ITI TF-2<sup>2</sup>, Appendix V as transport protocol. The XACML data-flow model serves as the underlying processing model.

The CH:ADR profile defines two actors and one transaction:

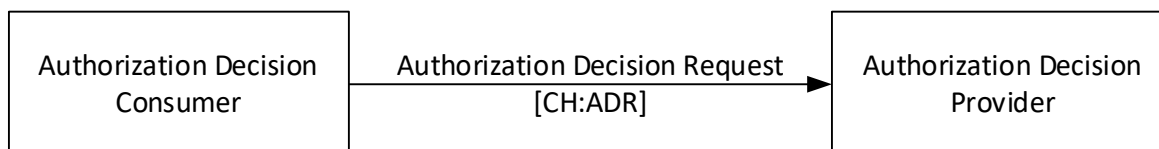


Figure 1: CH:ADR actor diagram

|        |                                                         |
|--------|---------------------------------------------------------|
| Actor: | CH:ADR Authorization Decision Provider                  |
| Role:  | This actor computes authorization decisions             |
| Actor: | CH:ADR Authorization Decision Consumer                  |
| Role:  | This actor queries and enforces authorization decisions |

Table 1: CH:ADR actors and roles

<sup>2</sup> IHE IT Infrastructure (ITI) Technical Framework, Volume 2, Revision 20.0, August 4, 2023.

Table 2 lists the transactions for each actor directly involved in the CH:ADR Profile. To claim compliance with this profile, an actor shall support all required transactions (labeled "R") and may support the optional transactions (labeled "O").

| Actors                                 | Transactions                            | Optionality | Section |
|----------------------------------------|-----------------------------------------|-------------|---------|
| CH:ADR Authorization Decision Consumer | Authorization Decision Request [CH:ADR] | R           | 3.1     |
| CH:ADR Authorization Decision Provider | Authorization Decision Request [CH:ADR] | R           | 3.1     |

Table 2: CH:ADR actors and transactions

Besides privacy policies created by a patient, an essential role for making authorization decisions is played by the CH:XUA assertion or CH:IUA token provided by the requesting user. Therefore, a CH:ADR Authorization Decision Consumer SHALL be able to obtain this assertion and forward it to the CH:ADR Authorization Decision Provider. This implies the groupings shown in Table 3:

| Actors                                 | Actor to be grouped with | Opt.                | Reference                                                                                                        |
|----------------------------------------|--------------------------|---------------------|------------------------------------------------------------------------------------------------------------------|
| CH:ADR Authorization Decision Consumer | XUA X-Service User       | O ( <i>Note 1</i> ) | Amendment 1 of Annex 5 EPRO-FDHA, Section 1.6                                                                    |
| CH:ADR Authorization Decision Consumer | IUA Authorization Client | O ( <i>Note 1</i> ) | Internet User Authorization (IUA), CH EPR FHIR (R4) <sup>3</sup> Implementation Guide, Version 5.0.0, 2025-12-18 |
| CH:ADR Authorization Decision Provider | XUA X-Service Provider   | R                   | Amendment 1 of Annex 5 EPRO-FDHA, Section 1.6                                                                    |
| CH:ADR Authorization Decision Provider | IUA Resource Server      | R                   | Internet User Authorization (IUA), CH EPR FHIR (R4) <sup>3</sup> Implementation Guide, Version 5.0.0, 2025-12-18 |

Table 3: CH:ADR required actors groupings

**Note 1:** The CH:ADR Authorization Decision Consumer SHALL at least be grouped with a XUA X-Service User or an IUA Authorization Client.

The actors "CH:ADR Authorization Decision Consumer" and "CH:ADR Authorization Decision Provider" can be seen as EPR-specific implementations respectively of Policy Enforcement Point (PEP) and Policy Decision Point (PDP) as defined in the XACML specification. The CH:PPQ or CH:PPQm Policy Repository acts as an XACML Policy Administration Point (PAP). The CH:ADR Authorization Decision Provider MAY use the CH:PPQ or CH:PPQm profile or any other mechanism to retrieve policies from the CH:PPQ or CH:PPQm Policy Repository.

Privacy policies of a patient are stored only in the reference community of this patient. Therefore, if the community where the CH:ADR Authorization Decision Consumer operates is not the reference community of the patient whose EPR the user wants to access, then the CH:ADR Authorization Decision Consumer SHALL query CH:ADR Authorization Decision Providers of other communities until one of them turns out to be the patient's reference community.

The following actors are grouped with the Actors XUA X-Service Provider, IUA Resource Server and CH:ADR Authorization Decision Consumer, and therefore called "Policy Enforcing Service Providers":

- l. XDS Document Registry;
- m. MHD Document Recipient;
- n. MHD Document Responder;
- o. CH:PPQ Policy Repository;
- p. CH:PPQm Policy Repository;

<sup>3</sup> The national extensions of the FHIR standard can be accessed free of charge at [fhir.ch](http://fhir.ch).

- q. CH:ATC Patient Audit Record Repository;
- r. RMU Update Responder.

All other actors grouped with XUA X-Service Provider or IUA Resource Server use indirect mechanisms to obtain authorization decisions whenever necessary – for example, by issuing a Registry Stored Query [ITI-18] and in this way delegating the CH:ADR Authorization Decision Consumer function to the XDS Document Registry. In other words, every CH:ADR Authorization Decision Consumer is a XUA X-Service Provider or an IUA Resource Server, but not every XUA X-Service Provider or IUA Resource Server is a CH:ADR Authorization Decision Consumer. The figures below show the correspondence between these actors, see also section 3.1.7 for details.

Figure 2 shows the Actors using XUA and CH:ADR to restrict access to sensitive data:

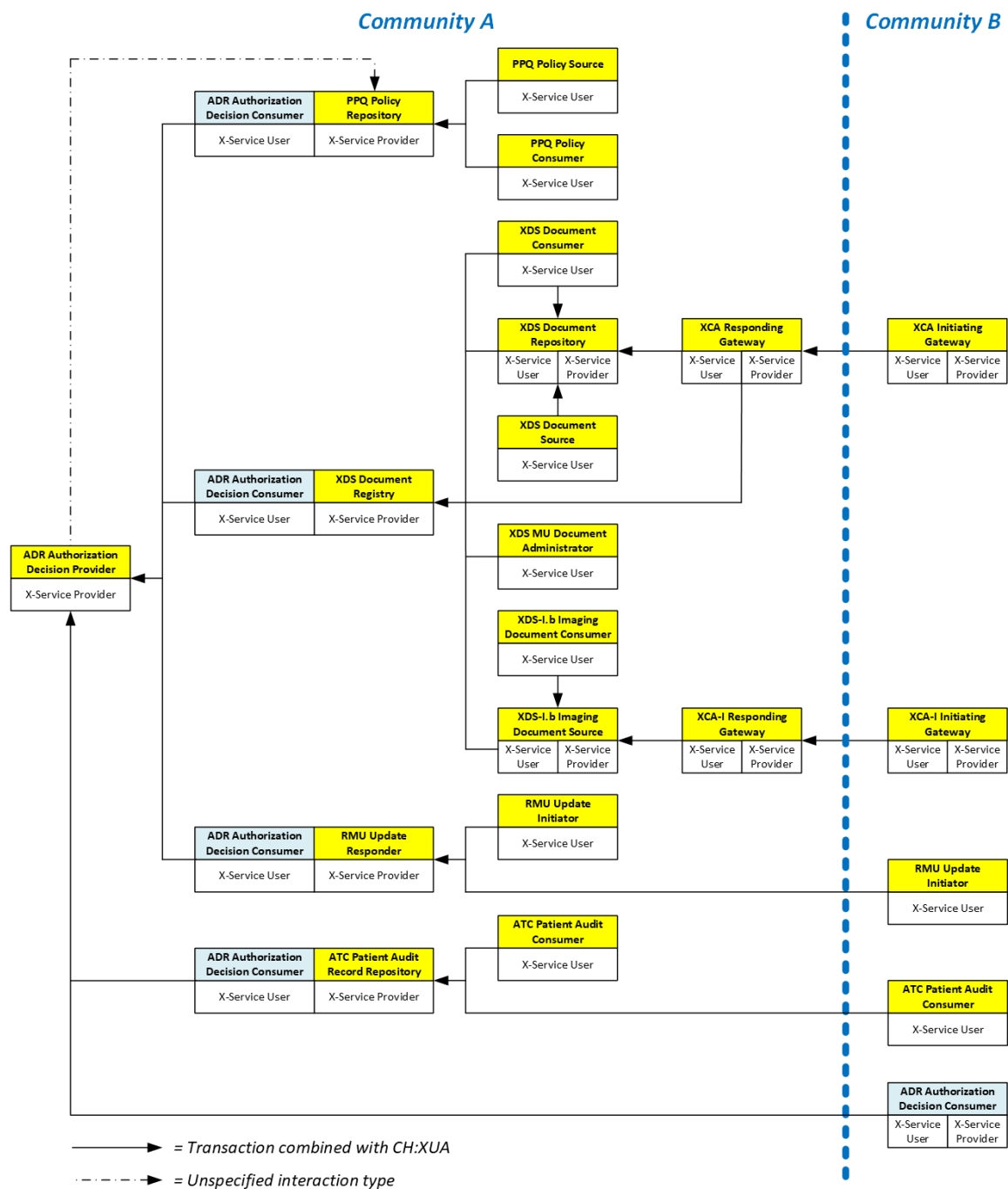


Figure 2: Actors using XUA and CH:ADR to restrict access to sensitive data

Figure 3 shows the Actors using IUA and CH:ADR to restrict access to sensitive data:

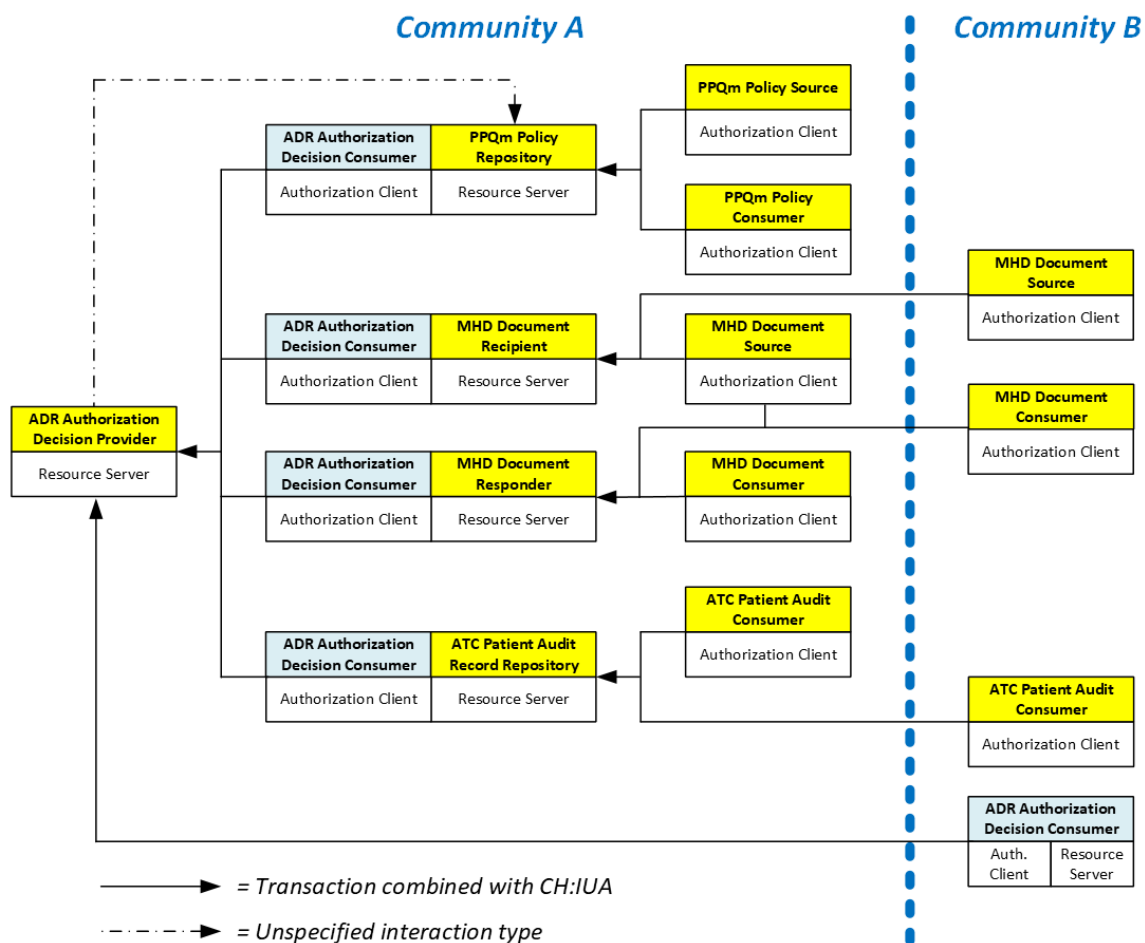


Figure 3: Actors using IUA and CH:ADR to access to sensitive data

## 2.2 Privacy Policy Query [CH:PPQ]

According to Swiss EPR regulations, patients have the right to decide who is allowed to access and modify data in their EPR and under which circumstance (cf. emergency access). The reference community of the patient is responsible for storing and considering these decisions.

For that, the EPR specifications prescribe to use XACML 2.0 format, and define the CH:PPQ Policy Repository as an XACML Policy Administration Point (PAP) that holds privacy policies of patients on-boarded in the given reference community. Each reference community provides a patient portal that allows patients to manage their privacy policies<sup>4</sup>.

The Privacy Policy Query (CH:PPQ) profile defines a programmatic interface to the CH:PPQ Policy Repository. Its transactions allow to add, query, update, and delete XACML policies and policy sets. In that way, this profile is somehow similar to XDS, but operates on another storage and another type of information. Same as XDS, CH:PPQ mandates the usage of CH:XUA.

CH:PPQ defines three actors and two transactions between them:

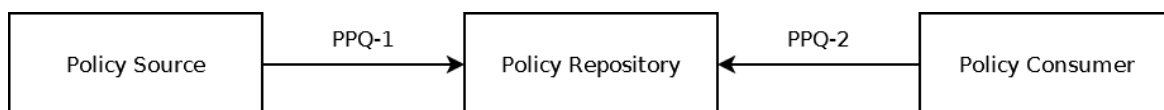


Figure 4: Privacy Policy Query (CH:PPQ) actor diagram

<sup>4</sup> XACML distinguishes between "policies" and "policy sets". In the given document, both will be referred to as "policies", unless explicitly stated otherwise.

While the CH:PPQ Policy Repository constitutes a specific implementation of a XACML Policy Administration Point (PAP), for the two other actors there is no correspondence in XACML. In this way, the CH:PPQ Policy Source and CH:PPQ Policy Consumer are introduced as entirely new actors.

The transactions between the profile's actors use SAML 2.0 profile of XACML v2.0 with EPR-specific extensions as data format and Web Services according to IHE ITI TF-2<sup>5</sup>, Appendix V as transport protocol.

|        |                                                                                                               |
|--------|---------------------------------------------------------------------------------------------------------------|
| Actor: | CH:PPQ Policy Repository                                                                                      |
| Role:  | This actor stores policies and policy sets and provides the possibility to add, query, update and delete them |
| Actor: | CH:PPQ Policy Source                                                                                          |
| Role:  | This actor initiates addition, update and deletion of policies and policy sets                                |
| Actor: | CH:PPQ Policy Consumer                                                                                        |
| Role:  | This actor queries policies and policy sets                                                                   |

Table 4: CH:PPQ actors and roles

Table 5 lists the transactions for each actor directly involved in the CH:PPQ Profile. To claim compliance with this profile, an actor shall support all required transactions (labeled "R") and may support the optional transactions (labeled "O").

| Actors                   | Transactions                       | Optionality | Section |
|--------------------------|------------------------------------|-------------|---------|
| CH:PPQ Policy Source     | Privacy Policy Feed [CH:PPQ-1]     | R           | 3.2     |
| CH:PPQ Policy Repository | Privacy Policy Feed [CH:PPQ-1]     | R           | 3.2     |
|                          | Privacy Policy Retrieve [CH:PPQ-2] | R           | 0       |
| CH:PPQ Policy Consumer   | Privacy Policy Retrieve [CH:PPQ-2] | R           | 0       |

Table 5: CH:PPQ actors and transactions

| Actors                   | Actor to be grouped with               | Reference                                       |
|--------------------------|----------------------------------------|-------------------------------------------------|
| CH:PPQ Policy Source     | XUA X-Service User                     | Amendment 1 of Annex 5 EPRO-FDHA, Section 1.6   |
| CH:PPQ Policy Repository | XUA X-Service Provider                 | Amendment 1 of Annex 5 EPRO-FDHA, Section 1.6   |
|                          | CH:ADR Authorization Decision Consumer | Amendment 2.1 of Annex 5 EPRO-FDHA, Section 2.1 |
| CH:PPQ Policy Consumer   | XUA X-Service User                     | Amendment 1 of Annex 5 EPRO-FDHA, Section 1.6   |

Table 6: CH:PPQ required actor groupings

Rules on authoring policies and policy sets are given in section 4.

<sup>5</sup> IHE IT Infrastructure (ITI) Technical Framework, Volume 2, Revision 20.0, August 4, 2023.

## 3 Volume 2 – Transactions

### 3.1 Authorization Decision Request [CH:ADR]

#### 3.1.1 Scope

This transaction is used by the CH:ADR Authorization Decision Consumer to query for authorization decisions, computed by the CH:ADR Authorization Decision Provider.

This transaction is based on synchronous Web services as described in IHE ITI TF-2<sup>6</sup>, Appendix V.

#### 3.1.2 Referenced Standards

- a. IHE ITI TF-2<sup>6</sup>, Appendix V "Web Services for IHE transactions"  
<https://profiles.ihe.net/ITI/TF/Volume2/ch-V.html>
- b. Assertions and Protocols for the OASIS Security Assertion Markup Language (SAML) V2.0  
<https://docs.oasis-open.org/security/saml/v2.0/saml-core-2.0-os.pdf>
- c. The home page of the "OASIS eXtensible Access Control Markup Language" technical committee: [https://www.oasis-open.org/committees/tc\\_home.php?wg\\_abbrev=xacml](https://www.oasis-open.org/committees/tc_home.php?wg_abbrev=xacml) references all XACML related protocols and specifications for implementers of this profile.
  - I. OASIS Multiple Resource Profile of XACML v2.0
  - II. [https://docs.oasis-open.org/xacml/2.0/access\\_control-xacml-2.0-mult-profile-spec-os.pdf](https://docs.oasis-open.org/xacml/2.0/access_control-xacml-2.0-mult-profile-spec-os.pdf)
  - III. OASIS Cross-Enterprise Security and Privacy Authorization (XSPA) Profile of SAML v2.0 for Healthcare Version 2.0 (not normative)
  - IV. <https://docs.oasis-open.org/xspa/saml-xspa/v2.0/saml-xspa-v2.0.html>
  - V. OASIS eXtensible Access Control Markup Language (XACML) v2.0
  - VI. (Original: [https://docs.oasis-open.org/xacml/2.0/access\\_control-xacml-2.0-core-spec-os.pdf](https://docs.oasis-open.org/xacml/2.0/access_control-xacml-2.0-core-spec-os.pdf)) Please be aware of the errata of the specification document as published on the XACML technical committee home page:
  - VII. **Errata:** [http://www.oasis-open.org/committees/download.php/24548/access\\_control-xacml-2.0-core-spec-os-errata.zip](http://www.oasis-open.org/committees/download.php/24548/access_control-xacml-2.0-core-spec-os-errata.zip) (spec and schema)
  - VIII. OASIS SAML 2.0 profile of XACML v2.0
  - IX. (Original: [http://docs.oasis-open.org/xacml/2.0/access\\_control-xacml-2.0-core-spec-os.pdf](http://docs.oasis-open.org/xacml/2.0/access_control-xacml-2.0-core-spec-os.pdf)) Please be aware of the errata of the specification document as published on the XACML technical committee home page:
  - X. **Errata:** [www.oasis-open.org/committees/download.php/24681/xacml-profile-saml2.0-v2-spec-wd-5-en.pdf](http://www.oasis-open.org/committees/download.php/24681/xacml-profile-saml2.0-v2-spec-wd-5-en.pdf)

<sup>6</sup> IHE IT Infrastructure (ITI) Technical Framework, Volume 2, Revision 20.0, August 4, 2023.

### 3.1.3 XML Namespaces

In addition to XML namespaces defined in IHE ITI TF-2<sup>7</sup>, Appendix V.2.4, the following namespaces and namespace prefixes will be used in message definitions:

| Prefix        | Namespace                                                        | Specification                                                                          |
|---------------|------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| saml          | urn:oasis:names:tc:SAML:2.0:assertion                            | Assertions and Protocols for the OA-SIS Security Assertion Markup Language (SAML) V2.0 |
| samlp         | urn:oasis:names:tc:SAML:2.0:protocol                             | Assertions and Protocols for the OA-SIS Security Assertion Markup Language (SAML) V2.0 |
| xacml-context | urn:oasis:names:tc:xacml:2.0:context:schema:os                   | OASIS eXtensible Access Control Markup Language (XACML) v2.0                           |
| xacml-saml    | urn:oasis:names:tc:xacml:2.0:profile:saml2.0:v2:schema:assertion | OASIS SAML 2.0 profile of XACML v2.0                                                   |
| xacml-samlp   | urn:oasis:names:tc:xacml:2.0:profile:saml2.0:v2:schema:protocol  | OASIS SAML 2.0 profile of XACML v2.0                                                   |

Table 7: Trigger events of CH:ADR request

### 3.1.4 Interaction Diagram

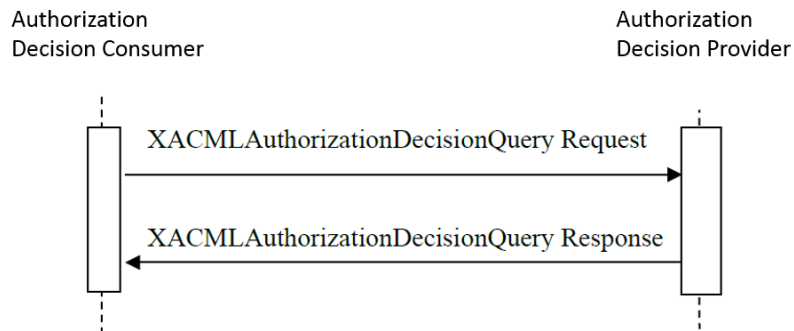


Figure 5: Sequence diagram of the CH:ADR transaction of the CH:ADR profile

<sup>7</sup> IHE IT Infrastructure (ITI) Technical Framework, Volume 2, Revision 20.0, August 4, 2023.

### 3.1.5 CH:ADR Request

The CH:ADR Authorization Decision Consumer uses this message to query the CH:ADR Authorization Decision Provider for authorization decisions. The request is an XML document compliant with the SAML v2.0 profile for XACML and contains information about which Subject (user) wants to perform which Action on which Resources. Multiple resources MAY be referenced in a single request as defined in the Multiple Resource Profile of XACML v2.0.

#### 3.1.5.1 Trigger Events

The CH:ADR Authorization Decision Consumer sends this message when it receives a request from a XUA X-Service User or IUA Authorization Client and needs to determine whether the requesting user is allowed to access the requested resources. The trigger events are summarized in Table 8:

| Trigger Event | Transaction                                  | Client                                     | Policy Enforcing Service Provider      |
|---------------|----------------------------------------------|--------------------------------------------|----------------------------------------|
| Documents     | Registry Stored Query [ITI-18]               | XDS Document Consumer <sup>8</sup>         | XDS Document Registry                  |
|               | Register Document Set-b [ITI-42]             | XDS Document Repository <sup>9</sup>       |                                        |
|               | Update Document Set [ITI-57]                 | XDS Metadata Update Document Administrator |                                        |
|               | Provide Document Bundle [ITI-65]             | MHD Document Source                        | MHD Document Recipient                 |
|               | Find Document References [ITI-67]            | MHD Document Consumer                      | MHD Document Responder                 |
|               | Retrieve Document [ITI-68]                   |                                            |                                        |
|               | Update Document Metadata [CH:MHD-1]          | MHD Document Source                        |                                        |
|               | Restricted Update Document Set [ITI-92]      | RMU Update Initiator                       | RMU Update Responder                   |
| Audit Trail   | Retrieve ATNA Audit Event [ITI-81]           | CH:ATC Patient Audit Consumer              | CH:ATC Patient Audit Record Repository |
| Policies      | Privacy Policy Feed [CH:PPQ-1]               | CH:PPQ Policy Source                       | CH:PPQ Policy Repository               |
|               | Privacy Policy Retrieve [CH:PPQ-2]           | CH:PPQ Policy Consumer                     |                                        |
|               | Mobile Privacy Policy Feed [CH:PPQ-3]        | CH:PPQm Policy Source                      | CH:PPQm Policy Repository              |
|               | Mobile Privacy Policy Bundle Feed [CH:PPQ-4] |                                            |                                        |
|               | Mobile Privacy Policy Retrieve [CH:PPQ-5]    | CH:PPQm Policy Consumer                    |                                        |

Table 8: Trigger events of CH:ADR request

<sup>8</sup> As Document Repositories and Imaging Document Sources are not grouped with an CH:ADR Authorization Decision Consumer, they use this transaction to indirectly query authorization decisions on data retrieval requests Retrieve Document Set [ITI-43] and Retrieve Imaging Document Set [RAD-69].

<sup>9</sup> As Document Repositories are not grouped with an CH:ADR Authorization Decision Consumer, they use this transaction to indirectly query authorization decisions on data submission requests Provide and Register Document Set-b [ITI-41] and Provide & Register Imaging Document Set – MTOM/XOP [RAD-68].

### 3.1.5.2 Message Semantics

The CH:ADR request message SHALL use SOAP 1.2 encoding and comply with all requirements described in IHE ITI TF-2<sup>10</sup>, Appendix V "Web Services for IHE Transaction".

The WS-Addressing header <wsa:Action> SHALL contain the value

`urn:e-health-suisse:2015:policy-enforcement:AuthorizationDecisionRequest`

The WS-Security header <wse:Security> SHALL contain a CH:XUA assertion or IUA token.

If IUA is used, the IUA token SHALL be contained in the <BinarySecurityToken> element of the Web Service security header with the ValueType attribute set to `urn:e-health-suisse:2021:JWT-Identity-Token` and the Encoding Type attribute set to `Base64Binary`.

The SOAP body SHALL contain a single element <xacml-samlp:XACMLAuthzDecisionQuery> with the following attributes:

- a. @ReturnContext SHOULD be set to "false", because the content of the CH:ADR request is not needed within the Authorization Result.
- b. @InputContextOnly SHALL be set to "false", as the CH:ADR Authorization Decision Provider may have further information and rules, other than the parameters included in the request, to determine a decision. This should not be restricted by the CH:ADR Authorization Decision Consumer.

CH:ADR does not define further constraints for other attributes of this element (see OASIS SAML 2.0 profile of XACML v2.0 for details).

The only child element of <xacml-samlp:XACMLAuthzDecisionQuery> SHALL be <xacml-context:Request> containing the following elements:

- c. Exactly one element <xacml-context:Subject>;
- d. One or more elements <xacml-context:Resource>;
- e. Exactly one element <xacml-context:Action>;
- f. Exactly one element <xacml-context:Environment>.

Their contents will be described in separate sub-sections below, but they have in common that they all represent lists of child elements <xacml-context:Attribute>. Each of those child elements is characterized by XML attributes @AttributeId and @DataType, and carries values in elements <xacml-context:AttributeValue>.

In each case, only required contents (i.e. required elements <xacml-context:Attribute>) will be described; CH:ADR Authorization Decision Consumers MAY provide any additional contents, CH:ADR Authorization Decision Providers MAY process or ignore them.

<sup>10</sup> IHE IT Infrastructure (ITI) Technical Framework, Volume 2, Revision 20.0, August 4, 2023.

### 3.1.5.2.1 Element <xacml-context:Subject>

This element identifies the requesting user. Its contents do not depend on the trigger event and represent a mapping of the CH:XUA assertion or IUA token.

The following child elements <xacml-context:Attribute> SHALL be provided:

| @AttributeId                                              | @DataType                               | Contents of <xacml-context:AttributeValue>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------------------------------------------|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| urn:oasis:names:tc:xacml:1.0:subject:subject-id           | http://www.w3.org/2001/XMLSchema#string | Text contents of the CH:XUA assertion element /Assertion/Subject/NameID or the JWT Claim subject_name in the CH:IUA token.                                                                                                                                                                                                                                                                                                                                                                                                                          |
| urn:oasis:names:tc:xacml:1.0:subject:subject-id-qualifier | http://www.w3.org/2001/XMLSchema#string | Contents of the CH:XUA assertion attribute /Assertion/Subject/NameID /@NameQualifier or the JWT Claim user_id_qualifier in the CH:IUA token.                                                                                                                                                                                                                                                                                                                                                                                                        |
| urn:ihe:iti:xca:2010:homeCommunityId                      | http://www.w3.org/2001/XMLSchema#anyURI | Text contents of the CH:XUA assertion element /Assertion/AttributeStatement /Attribute[@name="urn:ihe:iti:xca:2010:homeCommunityId"]/AttributeValue or the JWT Claim home_community_id in the CH:IUA token.                                                                                                                                                                                                                                                                                                                                         |
| urn:oasis:names:tc:xacml:2.0:subject:role                 | urn:hl7-org:v3#CV                       | Element <hl7:CodedValue> with all attributes (except @xsi:type) copied from the CH:XUA assertion element /Assertion/AttributeStatement /Attribute[@name="urn:oasis:names:tc:xacml:2.0:subject:role"]/AttributeValue/hl7:Role or the JWT Claim subject_role in the CH:IUA token.                                                                                                                                                                                                                                                                     |
| urn:oasis:names:tc:xspa:1.0:subject:organization-id       | http://www.w3.org/2001/XMLSchema#anyURI | Text contents of the CH:XUA assertion element /Assertion/AttributeStatement /Attribute[@name="urn:oasis:names:tc:xspa:1.0:subject:organization-id"]/AttributeValue or the JWT Claim subject_organization_id in the CH:IUA token.<br><br>As there can be multiple organization IDs in a CH:XUA assertion or a CH:IUA token, implementers SHALL map them either to separate elements <xacml-context:Attribute> with the same @AttributeId, or to multiple repetitions of <xacml-context:AttributeValue> in the same single <xacml-context:Attribute>. |
| urn:oasis:names:tc:xspa:1.0:subject:purposeofuse          | urn:hl7-org:v3#CV                       | Element <hl7:CodedValue> with all attributes (except @xsi:type) copied from the CH:XUA assertion element /Assertion/AttributeStatement /Attribute[@name="urn:oasis:names:tc:xspa:1.0:subject:purposeofuse"]/AttributeValue/hl7:PurposeOfUse or the JWT Claim purpose_of_use in the CH:IUA token.                                                                                                                                                                                                                                                    |

Table 9: Required attributes of the element <xacml-context:Subject>

### 3.1.5.2.2 Element <xacml-context:Resource>

This element depends on the trigger event (section 3.2.5.1) and identifies a particular object (for trigger events Policies and Audit Trail) or a class of objects (trigger event Documents) to which the requesting user tries to access.

## 3.1.5.2.2.1 Trigger event Policies

For the trigger event Policies, a separate repetition of the element `<xacml-context:Resource>` SHALL be provided for each policy or policy set being tried to be created, updated, read or deleted in a CH:PPQ or CH:PPQm request.

In each repetition, the following child elements `<xacml-context:Attribute>` SHALL be provided:

| @AttributeId                                                     | @DataType                               | Contents of <xacml-context:AttributeValue>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------------------------------------------------------------|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| urn:oasis:names:tc:xacml:1.0:resource:resource-id                | http://www.w3.org/2001/XMLSchema#anyURI | ID of the policy or policy set being tried to be created, updated, read or deleted                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| urn:e-health-suisse:2015:epr-spuid                               | urn:hl7-org:v3#II                       | Element <code>&lt;hl7:InstanceIdentifier&gt;</code> with attribute <code>@root</code> equal to the field CX.4.2 and attribute <code>@extension</code> equal to the field CX.1 of the CH:XUA assertion element <code>/Assertion/AttributeStatement/Attribute[@name="urn:oasis:names:tc:xacml:2.0:resource:resource-id"]/AttributeValue</code> or the JWT Claim <code>person_id</code> in the CH:IUA token.                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| urn:e-health-suisse:2015:policy-attributes:referenced-policy-set | http://www.w3.org/2001/XMLSchema#anyURI | ID of the policy or policy set referenced from the Policy Set being tried to be queried, added, updated or deleted. It is used in particular to ensure that a healthcare professional can only delegate access rights which do not exceed his own access level granted by the patient.<br><br>Example: A user tries to add a policy set with ID <code>c969c7cd-9fe9-4fdc-83c5-a7b5118922a3</code> (as in Resource attribute with <code>@AttributeId="urn:oasis:names:tc:xacml:1.0:resource:resource-id"</code> ) that puts a healthcare professional onto exclusion list, i.e. this policy set contains a reference to the policy set with the ID <code>"urn:e-health-suisse:2015:policies:exclusion-list"</code> . That is the value to be populated in the Resource attribute with <code>@AttributeId="urn:e-health-suisse:2015:policy-attributes:referenced-policy-set"</code> . |

Table 10: Required attributes of the element `<xacml-context:Resource>` for the trigger event Policies

The following additional child elements `<xacml-context:Attribute>` SHALL be provided for CH:PPQ or CH:PPQm request types `AddPolicy` and `UpdatePolicy`, if the policy set specifies an end date:

| @AttributeId                                          | @DataType                             | Contents of <xacml-context:AttributeValue>                              |
|-------------------------------------------------------|---------------------------------------|-------------------------------------------------------------------------|
| urn:e-health-suisse:2023:policy-attributes:start-date | http://www.w3.org/2001/XMLSchema#date | Start date, if the policy set specifies it, the current date otherwise. |
| urn:e-health-suisse:2023:policy-attributes:end-date   | http://www.w3.org/2001/XMLSchema#date | Policy set's end date.                                                  |

Table 10a: Conditionally required attributes of the element `<xacml-context:Resource>` for the trigger event Policies

## 3.1.5.2.2.2 Trigger event Documents

For the trigger event Documents, exactly three repetitions of the element `<xacml-context:Resource>` SHALL be provided — one for each possible confidentiality level of documents in EPR (normal, restricted, secret).

In each repetition, the following child elements `<xacml-context:Attribute>` SHALL be provided:

| @AttributeId                                      | @DataType                               | Contents of <xacml-context:AttributeValue>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------------------------------|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| urn:oasis:names:tc:xacml:1.0:resource:resource-id | http://www.w3.org/2001/XMLSchema#anyURI | <p>A string of the form "urn:e-health-suisse:2015:epr-subset:&lt;epr-spuid&gt;:&lt;conf-level&gt;", where:</p> <ul style="list-style-type: none"> <li>&lt;epd-spuid&gt; is the EPR-SPID of the patient,</li> <li>&lt;conf-level&gt; is the confidentiality level name corresponding to the given repetition of <code>&lt;xacml-context:Resource&gt;</code>, i.e. "normal", "restricted", or "secret".</li> </ul> <p>For example, for a patient with the EPR-SPID "8901", <code>&lt;xacml-context:Resource&gt;</code> repetitions with the following attribute values will have to be provided:</p> <ul style="list-style-type: none"> <li>urn:e-health-suisse:2015:epr-subset:8901:normal</li> <li>urn:e-health-suisse:2015:epr-subset:8901:restricted</li> <li>urn:e-health-suisse:2015:epr-subset:8901:secret</li> </ul> |
| urn:e-health-suisse:2015:epr-spuid                | urn:hl7-org:v3#II                       | <p>Element <code>&lt;hl7:InstanceIdentifier&gt;</code> with attribute <code>@root</code> equal to the field CX.4.2 and attribute <code>@extension</code> equal to the field CX.1 of the CH:XUA assertion element <code>/Assertion/AttributeStatement/Attribute[@name="urn:oasis:names:tc:xacml:2.0:resource:resource-id"]/AttributeValue</code> or the JWT Claim <code>person_id</code> in the CH:IUA token.</p>                                                                                                                                                                                                                                                                                                                                                                                                           |
| urn:ihe:iti:xds-b:2007:confidentiality-code       | urn:hl7-org:v3#CV                       | <p>Confidentiality level code corresponding to the given repetition of <code>&lt;xacml-context:Resource&gt;</code>, in HL7v3 Coded Value format, e.g.:</p> <pre>&lt;hl7:CodedValue code="17621005"   codeSystem=     "2.16.840.1.113883.6.96"   displayName="normal"/&gt;</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| urn:ihe:iti:xca:2010:homeCommunityId              | http://www.w3.org/2001/XMLSchema#anyURI | OID of the CH:ADR Authorization Decision Consumer's community in the URN format                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

Table 11: Required attributes of the element `<xacml-context:Resource>` for the trigger event Documents

## 3.1.5.2.2.3 Trigger event Audit Trail

For the trigger event Audit Trail, exactly one repetition of the element `<xacml-context:Resource>` SHALL be provided.

The following child elements `<xacml-context:Attribute>` SHALL be provided:

| @AttributeId                                      | @DataType                               | Contents of <xacml-context:AttributeValue>                                                                                                                                                                                                                                                                                                                                                                             |
|---------------------------------------------------|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| urn:oasis:names:tc:xacml:1.0:resource:resource-id | http://www.w3.org/2001/XMLSchema#anyURI | A string of the form "urn:e-health-suisse:2015:epr-subset:<epr-sp-id>:patient-audit-trail-records", where <epr-sp-id> is the EPR-SPID of the patient specified in the query parameter "entity-id".<br>For example, for a patient with the EPR-SPID "8901", the following value will be created: <ul style="list-style-type: none"> <li>urn:e-health-suisse:2015:epr-subset:8901:patient-audit-trail-records</li> </ul> |
| urn:e-health-suisse:2015:epr-sp-id                | urn:hl7-org:v3#II                       | Element <code>&lt;hl7:InstanceIdentifier&gt;</code> with attribute <code>@root</code> equal to the field CX.4.2 and attribute <code>@extension</code> equal to the field CX.1 of the CH:XUA assertion element <code>/Assertion/AttributeStatement/Attribute[@name="urn:oasis:names:tc:xacml:2.0:resource:resource-id"]/AttributeValue</code> or the JWT Claim <code>person_id</code> in the CH:IUA token.              |

Table 12: Required attributes of the element `<xacml-context:Resource>` for the trigger event CH:ATC

3.1.5.2.3 Element `<xacml-context:Action>`

This element identifies the action the requesting user tries to perform.

The following child element `<xacml-context:Attribute>` SHALL be provided:

- a. `@AttributeId`: fixed value "urn:oasis:names:tc:xacml:1.0:action:action-id"
- b. `@DataType`: fixed value "http://www.w3.org/2001/XMLSchema#anyURI"
- c. Contents of `<xacml-context:AttributeValue>`: fixed value depending from the trigger transaction (see Table 8):
  - I. For ITI-18, ITI-67 and ITI-68 – urn:ihe:iti:2007:RegistryStoredQuery
  - II. For ITI-42 and ITI-65 – urn:ihe:iti:2007:RegisterDocumentSet-b
  - III. For ITI-57 – urn:ihe:iti:2010:UpdateDocumentSet
  - IV. For ITI-81 – urn:e-health-suisse:2015:patient-audit-administration:RetrieveAtnaAudit
  - V. For ITI-92 and MHD-1 – urn:ihe:iti:2018:RestrictedUpdateDocumentSet
  - VI. For PPQ-1, PPQ-3 and PPQ-4 – one of:
    - i. urn:e-health-suisse:2015:policy-administration:AddPolicy
    - ii. urn:e-health-suisse:2015:policy-administration:UpdatePolicy
    - iii. urn:e-health-suisse:2015:policy-administration>DeletePolicy
  - VII. For PPQ-2 and PPQ-5 – urn:e-health-suisse:2015:policy-administration:PolicyQuery

3.1.5.2.4 Element `<xacml-context:Environment>`

The EPR does not specify any required attributes in the element `<xacml-context:Environment>` within `<xacml-sampl:XACMLAuthzDecisionQuery>`. Therefore, this child element MAY be empty. The CH:ADR Authorization Decision Provider MAY process or ignore any provided attributes.

### 3.1.5.3 Expected Actions

The CH:ADR Authorization Decision Provider SHALL compute and return authorization decisions that match the XACML Query parameters according to the rules defined in XACML policies.

As the entry point for the evaluation, the CH:ADR Authorization Decision Provider SHALL use:

- a. All available policy sets that reference the patient specified in the CH:ADR request (in the element `<xacml-context:Attribute>` with the attribute `@AttributeId` equal to "urn:e-health-suisse:2015:epr-spuid"), i.e. the ones produced from templates 201–203 and 301–304 of the official Policy Stack (see chapter 4).
- b. Patient-unspecific policy sets related to Document Administrator and Policy Administrator roles, i.e. policy sets 110 and 111 from the official Policy Stack.
- c. All other policies and policy sets are referenced transitively from the aforementioned ones.

The CH:ADR Authorization Decision Provider SHALL create a CH:ADR response message that conveys the results of the evaluation of the patient's privacy policies against the request. One result for each Resource SHALL be included in the response message.

### 3.1.6 CH:ADR Response

The CH:ADR response message is created by the CH:ADR Authorization Decision Provider in response to the CH:ADR request. This message conveys to the CH:ADR Authorization Decision Consumer the results of the evaluation made by the CH:ADR Authorization Decision Provider. For each Resource specified within the request message, the CH:ADR Authorization Decision Provider provides an authorization decision that SHALL be used by the CH:ADR Authorization Decision Consumer to determine which of the requested objects are to be returned or transactions to be allowed in response to the corresponding trigger transactions.

#### 3.1.6.1 Trigger Events

This message is created by the CH:ADR Authorization Decision Provider after the evaluation of the CH:ADR request message. The CH:ADR Authorization Decision Provider SHALL only return authorization decisions applicable to the request.

#### 3.1.6.2 Message Semantics

The CH:ADR response message SHALL use SOAP 1.2 encoding and comply with all requirements described in IHE ITI TF-2<sup>11</sup>, Appendix V "Web Services for IHE Transaction".

The WS-Addressing header `<wsa:Action>` SHALL contain the value

`urn:e-health-suisse:2015:policy-enforcement:XACMLAuthzDecisionResponse`

The body of the SOAP message SHALL contain a single element `<samlp:Response>`, which SHALL contain a single element `<samlp:Status>` and a single element `<saml:Assertion>`:

The element `<samlp:Status>` contains a single element `<samlp:StatusCode>`, whose attribute `@Value` SHALL be set as follows:

- a. If status codes of all decisions are "urn:e-health-suisse:2015:error:not-holder-of-patient-policies" (see below), `@Value` SHALL equal to "urn:e-health-suisse:2015:error:not-holder-of-patient-policies".
- b. Otherwise, `@Value` SHALL be set as defined in section 4.10 "Element `<samlp:Response>`: XACMLAuthzDecision Response" of OASIS SAML 2.0 profile of XACML v2.0 (errata).

The assertion in `<saml:Assertion>` SHALL contain following elements:

- c. Exactly one element `<saml:Issuer>` SHALL identify the CH:ADR Authorization Decision Provider by containing the home community ID of its community encoded as an URN and an XML attribute `@NameQualifier` equal to "urn:e-health-suisse:community-index".

<sup>11</sup> IHE IT Infrastructure (ITI) Technical Framework, Volume 2, Revision 20.0, August 4, 2023.

- d. Exactly one element <saml:Statement> of the type "xacml-saml:XACMLAuthzDecision-StatementType" with a single element <xacml-context:Response>.

As specified in the OASIS multiple resource profile of XACML v2.0, the <xacml-context:Response> element SHALL contain an element <xacml-context:Result> for each element <xacml-context:Resource> contained within the CH:ADR request message.

Contents of <xacml-context:Result> are the following:

- e. Attribute @Resourceid SHALL contain the resource ID of the corresponding <xacml-context:Resource> (provided there in the element <xacml-context:Attribute> with @Attributeld equal to "urn:oasis:names:tc:xacml:1.0:resource:resource-id").
- f. Child element <xacml-context:Decision> SHALL hold the actual decision code, i.e. one of:
  - I. "Permit": if the evaluation was successful and the Subject is authorized to perform the Action on the Resource;
  - II. "Deny": if the evaluation was successful and the Subject is explicitly prohibited to perform the Action on the Resource.
  - III. "NotApplicable": if the evaluation was successful, but the Subject is neither explicitly authorized nor explicitly prohibited to perform the Action on the Resource.
  - IV. "Indeterminate": either the evaluation of policies failed, or the policies are not available because the community of the CH:ADR Authorization Decision Provider is not the reference community of the given patient.
- g. Child element <xacml-context:Status> SHALL contain a element <xacml-context:StatusCode>, whose attribute @Value SHALL be set as follows:
  - I. When policies are not available because the community of the CH:ADR Authorization Decision Provider is not the reference community of the given patient, @Value SHALL equal to "urn:e-health-suisse:2015:error:not-holder-of-patient-policies".
  - II. Otherwise, @Value SHALL be set as defined in section B.9 "Status Codes" of OASIS eXtensible Access Control Markup Language (XACML) v2.0.

Other elements of <xacml-context:Status> MAY be optionally provided as well.

### 3.1.6.3 Expected Actions

On receiving a CH:ADR response message, the Policy Enforcing Service Provider SHALL enforce the decision as described below.

Note that Registry Stored Query [ITI-18], Find Document References [ITI-67], Privacy Policy Retrieve [CH:PPQ-2] and Mobile Privacy Policy Retrieve [CH:PPQ-5] are the only transactions allowing partial success; all others obey the principle of atomicity ("all or nothing").

If the decision is "Deny" or "NotApplicable":

| Transaction               | Effect                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ITI-18<br>ITI-67          | Documents with a confidentiality code for which such a decision was returned SHALL NOT be disclosed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| ITI-42<br>ITI-65          | <p><u>Deny:</u><br/>The whole request containing a document with the confidentiality code for which such a decision was returned, SHALL be rejected and an error message SHALL be returned to the client.</p> <p><u>NotApplicable:</u></p> <ol style="list-style-type: none"> <li>1. If this decision relates to the confidentiality level "Normal", and the same CH:ADR response contains a decision "Permit" for the confidentiality level "Restricted", then the confidentiality level SHALL be set to "Restricted" in all documents in the current submission request that have the confidentiality level "Normal".</li> <li>2. If this decision relates to the confidentiality levels "Normal" and "Restricted", and the same CH:ADR response contains a decision "Permit" for the confidentiality level "Secret", then the confidentiality level SHALL be set to "Secret" for all documents in the current submission request that have the confidentiality level "Normal" or "Restricted".</li> <li>3. If, after applying rules 1 and 2, all documents have confidentiality levels that are permitted by the CH:ADR response, then the behavior SHALL be as if the decision were "Permit".</li> <li>4. Otherwise, the behavior SHALL be as if the decision were "Deny".</li> </ol> |
| ITI-57<br>ITI-92<br>MHD-1 | If the request references a document, whose confidentiality code stored in the EPR platform (i.e. the current value and not the possibly different new value specified in the given metadata update request) is the one for which such a decision was returned, then the whole request SHALL be rejected and an error message SHALL be returned to the client.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| ITI-81                    | The CH:ATC Patient Audit Record repository SHALL NOT disclose the related patient audit records in the Retrieve ATNA Audit Event [ITI-81] response.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| PPQ-1<br>PPQ-3<br>PPQ-4   | The CH:PPQ or CH:PPQm Policy Repository SHALL reject the whole request containing (for AddPolicy and UpdatePolicy) or referencing (for DeletePolicy) a policy or policy set for which such a decision was returned and return an error message to the CH:PPQ or CH:PPQm Policy Source.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| PPQ-2<br>PPQ-5            | The CH:PPQ or CH:PPQm Policy Repository SHALL NOT disclose policies and policy sets for which such a decision was returned.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

Table 13: Effects of the authorization decisions "Deny" and "NotApplicable"

If the decision is "Permit":

| Transaction               | Effect                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ITI-18<br>ITI-67          | The following objects SHALL be disclosed:<br>a) document entries or document references with the confidentiality code for which such a decision was returned, if they reference the same patient as the CH:XUA assertion or IUA token;<br>b) submission sets, if they reference the same patient as the CH:XUA assertion or IUA token;<br>c) associations that relate to at least one object from subsets a) or b);<br>d) references to objects from subsets a), b), and c). |
| ITI-42<br>ITI-65          | The request SHALL be processed if the decision Permit was returned for all documents and the patient referenced in the CH:XUA assertion or IUA token is the same.                                                                                                                                                                                                                                                                                                            |
| ITI-57<br>ITI-92<br>MHD-1 | The request SHALL be processed if the decision Permit was returned for all current confidentiality codes of the documents referenced in the request and the patient is the same as in the CH:XUA assertion or IUA token.                                                                                                                                                                                                                                                     |
| ITI-81                    | The CH:ATC Patient Audit Record repository SHALL disclose related patient audit records in the Retrieve ATNA Audit Event [ITI-81] response if they reference the same patient as the CH:XUA assertion or IUA token.                                                                                                                                                                                                                                                          |
| PPQ-1<br>PPQ-3<br>PPQ-4   | The CH:PPQ or CH:PPQm Policy Repository SHALL process the request if the decision Permit was returned for all policy sets contained (for AddPolicy and UpdatePolicy) or referenced (for DeletePolicy) in the request and all these policy sets reference the same patient as the CH:XUA assertion or IUA token.                                                                                                                                                              |
| PPQ-2<br>PPQ-5            | The CH:PPQ or CH:PPQm Policy Repository SHALL disclose policies and policy sets for which such a decision was returned and which at the same time either do not reference any patient or reference the same patient as the CH:XUA assertion or IUA token.                                                                                                                                                                                                                    |

Table 14: Effects of the authorization decision "Permit"

If the decision is "Indeterminate":

| Transaction                                                                 | Effect                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ITI-18<br>ITI-42<br>ITI-57<br>ITI-65<br>ITI-67<br>ITI-81<br>ITI-92<br>MHD-1 | The CH:ADR Authorization Decision Consumer SHALL query CH:ADR Authorization Decision Providers of other communities (as listed in the Community Portal Index (CPI)), until a response contains a decision code other than "Indeterminate".<br>If all available CH:ADR Authorization Decision Providers return the decision "Indeterminate", then the CH:ADR Authorization Decision Consumer SHALL apply the same rules as if the decision were "Deny". |
| PPQ-1<br>PPQ-2<br>PPQ-3<br>PPQ-4<br>PPQ-5                                   | The CH:PPQ or CH:PPQm Policy Repository SHALL apply the same rules as if the decision were "Deny".                                                                                                                                                                                                                                                                                                                                                     |

Table 15: Effects of the authorization decision "Deny"

### 3.1.7 Indirect decision queries

#### 3.1.7.1 Decision querying and enforcement for Retrieve Document Set [ITI-43]

To determine whether the requesting user is allowed to retrieve documents referenced in a Retrieve Document Set [ITI-43] request, the XDS Document Repository SHALL create for each of these documents an Registry Stored Query [ITI-18] of the type "GetDocuments" and send this query to the XDS Document Registry along with the CH:XUA assertion obtained from the XDS Document Consumer.

If the corresponding Registry Stored Query response contains the entryUUID of the document, then the document SHALL be supplied to the XDS Document Consumer. If the corresponding Registry Stored Query Response does not contain the entryUUID of the document, then the document SHALL NOT be supplied to the Document Consumer.

#### 3.1.7.2 Decision querying and enforcement for Retrieve Imaging Document Set [RAD-69]

The retrieval of images SHALL be enforced according to the access rights formulated by the patient. If a Retrieve Imaging Document Set [RAD-69] request addresses a DICOM object, and the CH:XUA assertion contained in the request references a user and a patient, the DICOM object SHALL be delivered to the user only if the repositories of the community contain a KOS object which references the study instance id and the user is authorized to access the KOS object.

The Imaging Document Source actor SHALL perform the following steps when processing each element "StudyRequest" of a Retrieve Imaging Document Set [RAD-69] request:

In the XDS infrastructure:

- a. Query the XDS metadata of KOS objects of the DICOM study by sending a Registry Stored Query [ITI-18] of type "FindDocumentsByReferenceId" with the following parameters:
  - I. \$XDSDocumentEntryPatientId: MPI-PID of the patient.
  - II. \$XDSDocumentEntryStatus: fixed value "urn:oasis:names:tc:ebxml-regrep:StatusType:Approved".
  - III. \$XDSDocumentEntryFormatCode: fixed value "1.2.840.10008.5.1.4.1.1.88.59^^1.2.840.10008.2.6.1" (format code of KOS objects according to section 2.3 annex 3, in the HL7v2 CE format).
  - IV. \$XDSDocumentEntryReferenceIdList: StudyInstanceUID in the CXI format, e.g. "2.16.5.4.3.2.1.0^^^^urn:ihe:iti:xds:2016:studyInstanceUID".
- b. The CH:XUA assertion contained in the original Retrieve Imaging Document Set [RAD-69] request SHALL be used for this query.
- c. This query may return metadata of more than one KOS object. If no KOS objects have been found, then no DICOM contents SHALL be returned.
- d. Retrieve the contents of the found KOS objects by sending a Retrieve Document Set [ITI-43] query. The CH:XUA assertion contained in the original Retrieve Imaging Document Set [RAD-69] request SHALL be used for this query.
- e. Match the DICOM object references in the KOS object and in the Retrieve Imaging Document Set [RAD-69] request. DICOM objects not referenced in both, the KOS object and the Retrieve Imaging Document Set [RAD-69] request SHALL NOT be returned in the response.

### 3.1.8 Security Considerations

Relevant Security Considerations are defined in IHE ITI TF-1<sup>12</sup>, chapter 9.4. The CH:ADR profile requires all actors to be grouped with Secure Node or Secure Application implementing the "STX: TLS 1.2 floor using BCP195 Option" defined in the IHE ITI TF-2<sup>13</sup>, chapter 3.19.6.2.3. Relevant XDS

<sup>12</sup> IHE IT Infrastructure (ITI) Technical Framework, Volume 1, Revision 20.0, August 4, 2023.

<sup>13</sup> IHE IT Infrastructure (ITI) Technical Framework, Volume 2, Revision 20.0, August 4, 2023.

Affinity Domain Security background is discussed in the XDS Security Considerations section (IHE ITI TF-1<sup>12</sup>, chapter 10.7). The involved actors SHALL record audit events according to the following definitions:

### 3.1.8.1 CH:ADR Authorization Decision Consumer Audit Message

|                                                    | Field Name            | Opt | Value Constraints                                            |
|----------------------------------------------------|-----------------------|-----|--------------------------------------------------------------|
| <b>Event</b>                                       | EventID               | M   | EV (110112, DCM,"Query")                                     |
|                                                    | EventActionCode       | M   | E = Execute                                                  |
|                                                    | EventDateTime         | M   | <i>not specialized</i>                                       |
|                                                    | EventOutcomeIndicator | M   | <i>not specialized</i>                                       |
|                                                    | EventTypeCode         | M   | EV("ADR", "e-health-suisse", "Authorization Decision Query") |
| Source (Authorization Decision Consumer) (1)       |                       |     |                                                              |
| Human Requestor (0..n)                             |                       |     |                                                              |
| Destination (Authorization Decision Provider) (1)  |                       |     |                                                              |
| Audit Source (Authorization Decision Consumer) (1) |                       |     |                                                              |
| Requester Entity (1)                               |                       |     |                                                              |
| Resource (1..n)                                    |                       |     |                                                              |

Where:

|                                                             |                            |    |                                                                                      |
|-------------------------------------------------------------|----------------------------|----|--------------------------------------------------------------------------------------|
| <b>Source:<br/>AuditMessage/<br/>ActiveParticipant</b>      | UserID                     | U  | <i>not specialized</i>                                                               |
|                                                             | AlternativeUserID          | M  | the process ID as used within the local operating system in the local system of logs |
|                                                             | UserName                   | U  | <i>not specialized</i>                                                               |
|                                                             | UsersRequestor             | U  | <i>not specialized</i>                                                               |
|                                                             | RoleIDCode                 | M  | EV (110153, DCM, "Source Role ID")                                                   |
|                                                             | NetworkAccessPointTypeCode | M  | "1" for machine (DNS) name, "2" for IP address                                       |
|                                                             | NetworkAccessPointID       | M  | The machine name or IP address                                                       |
| <b>Human Requestor: AuditMessage/<br/>ActiveParticipant</b> | UserID                     | M  | Identity of the human that initiated the transaction.                                |
|                                                             | AlternativeUserID          | U  | <i>not specialized</i>                                                               |
|                                                             | UserName                   | U  | <i>not specialized</i>                                                               |
|                                                             | UsersRequestor             | U  | <i>not specialized</i>                                                               |
|                                                             | RoleIDCode                 | U  | Access Control role(s) the user holds that allows this transaction.                  |
|                                                             | NetworkAccessPointTypeCode | NA |                                                                                      |
|                                                             | NetworkAccessPointID       | NA |                                                                                      |
| <b>Destination:<br/>AuditMessage/<br/>ActiveParticipant</b> | UserID                     | M  | SOAP Endpoint URI                                                                    |
|                                                             | AlternativeUserID          | U  | <i>not specialized</i>                                                               |
|                                                             | UserName                   | U  | <i>not specialized</i>                                                               |
|                                                             | UsersRequestor             | U  | <i>not specialized</i>                                                               |
|                                                             | RoleIDCode                 | M  | EV (110152, DCM, "Destination Role ID")                                              |
|                                                             | NetworkAccessPointTypeCode | M  | "1" for machine (DNS) name, "2" for IP address                                       |
|                                                             | NetworkAccessPointID       | M  | The machine name or IP address                                                       |
| <b>Audit Source:<br/>AuditMessage/</b>                      | AlternativeUserID          | U  | <i>not specialized</i>                                                               |
|                                                             | UserName                   | U  | <i>not specialized</i>                                                               |

|                                                                                     |                                       |          |                                                                                                                                                                            |
|-------------------------------------------------------------------------------------|---------------------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>AuditSourceIdentification</b>                                                    | <i>UsersRequestor</i>                 | <i>U</i> | <i>not specialized</i>                                                                                                                                                     |
| <b>Requester Entity:<br/>AuditMessage/<br/>ParticipantObject<br/>Identification</b> | ParticipantObjectTypeCode             | M        | "1" (person)                                                                                                                                                               |
|                                                                                     | ParticipantObjectTypeCodeRole         | M        | "11" (security user entity)                                                                                                                                                |
|                                                                                     | <i>ParticipantObjectDataLifeCycle</i> | <i>U</i> | <i>not specialized</i>                                                                                                                                                     |
|                                                                                     | ParticipantObjectIDTypeCode           | M        | Subject Role (value of the attribute "urn:oasis:names:tc:xacml:2.0:subject:role")                                                                                          |
|                                                                                     | <i>ParticipantObjectSensitivity</i>   | <i>U</i> | <i>not specialized</i>                                                                                                                                                     |
|                                                                                     | ParticipantObjectID                   | M        | Subject ID (value of the attribute "urn:oasis:names:tc:xacml:1.0:subject:subject-id")                                                                                      |
|                                                                                     | <i>ParticipantObjectName</i>          | <i>U</i> | <i>not specialized</i>                                                                                                                                                     |
|                                                                                     | <i>ParticipantObjectQuery</i>         | <i>U</i> | <i>not specialized</i>                                                                                                                                                     |
|                                                                                     | <i>ParticipantObjectDetail</i>        | <i>U</i> | <i>not specialized</i>                                                                                                                                                     |
| <b>Resource:<br/>AuditMessage/<br/>ParticipantObject<br/>Identification</b>         | ParticipantObjectTypeCode             | M        | "2" (system)                                                                                                                                                               |
|                                                                                     | ParticipantObjectTypeCodeRole         | M        | For CH:ADR due to XDS, MHD or RMU:<br>"3" (report)<br>For CH:ADR due to CH:PPQ or CH:PPQm:<br>"13" (security resource)<br>For CH:ADR due to ATC:<br>"17" (data repository) |
|                                                                                     | <i>ParticipantObjectDataLifeCycle</i> | <i>U</i> | <i>not specialized</i>                                                                                                                                                     |
|                                                                                     | ParticipantObjectIDTypeCode           | <i>U</i> | <i>not specialized</i>                                                                                                                                                     |
|                                                                                     | <i>ParticipantObjectSensitivity</i>   | <i>U</i> | <i>not specialized</i>                                                                                                                                                     |
|                                                                                     | ParticipantObjectID                   | M        | Resource-ID (value of the attribute "urn:oasis:names:tc:xacml:1.0:resource:resource-id")                                                                                   |
|                                                                                     | <i>ParticipantObjectName</i>          | <i>U</i> | <i>not specialized</i>                                                                                                                                                     |
|                                                                                     | <i>ParticipantObjectQuery</i>         | <i>U</i> | <i>not specialized</i>                                                                                                                                                     |
|                                                                                     | <i>ParticipantObjectDetail</i>        | C        | Authorization result (when available):<br>type = fixed string "decision",<br>value = one of "Permit", "Deny", "NotApplicable", "Indeterminate".                            |

Table 16: CH:ADR Authorization Decision Consumer Audit Message

## 3.1.8.2 CH:ADR Authorization Decision Provider Audit Message

|                                                    | Field Name                   | Opt      | Value Constraints                                            |
|----------------------------------------------------|------------------------------|----------|--------------------------------------------------------------|
| <b>Event</b>                                       | EventID                      | M        | EV (110112, DCM,"Query")                                     |
|                                                    | EventActionCode              | M        | E = Execute                                                  |
|                                                    | <i>EventDateTime</i>         | <i>M</i> | <i>not specialized</i>                                       |
|                                                    | <i>EventOutcomeIndicator</i> | <i>M</i> | <i>not specialized</i>                                       |
|                                                    | EventTypeCode                | M        | EV("ADR", "e-health-suisse", "Authorization Decision Query") |
| Source (Authorization Decision Consumer) (1)       |                              |          |                                                              |
| Human Requestor (0..n)                             |                              |          |                                                              |
| Destination (Authorization Decision Provider) (1)  |                              |          |                                                              |
| Audit Source (Authorization Decision Provider) (1) |                              |          |                                                              |
| Requester Entity (1)                               |                              |          |                                                              |
| Resource (1..n)                                    |                              |          |                                                              |

Where:

|                                                                                     |                                       |           |                                                                                       |
|-------------------------------------------------------------------------------------|---------------------------------------|-----------|---------------------------------------------------------------------------------------|
| <b>Source:<br/>AuditMessage/<br/>ActiveParticipant</b>                              | <i>UserID</i>                         | <i>U</i>  | <i>not specialized</i>                                                                |
|                                                                                     | <i>AlternativeUserID</i>              | <i>U</i>  | <i>not specialized</i>                                                                |
|                                                                                     | <i>UserName</i>                       | <i>U</i>  | <i>not specialized</i>                                                                |
|                                                                                     | <i>UsersRequestor</i>                 | <i>U</i>  | <i>not specialized</i>                                                                |
|                                                                                     | <i>RoleIDCode</i>                     | <i>M</i>  | EV (110153, DCM, "Source Role ID")                                                    |
|                                                                                     | <i>NetworkAccessPointTypeCode</i>     | <i>U</i>  | "1" for machine (DNS) name,<br>"2" for IP address                                     |
|                                                                                     | <i>NetworkAccessPointID</i>           | <i>U</i>  | The machine name or IP address                                                        |
| <b>Human Requestor:<br/>AuditMessage/<br/>ActiveParticipant</b>                     | <i>UserID</i>                         | <i>M</i>  | Identity of the human that initiated the transaction.                                 |
|                                                                                     | <i>AlternativeUserID</i>              | <i>U</i>  | <i>not specialized</i>                                                                |
|                                                                                     | <i>UserName</i>                       | <i>U</i>  | <i>not specialized</i>                                                                |
|                                                                                     | <i>UsersRequestor</i>                 | <i>U</i>  | <i>not specialized</i>                                                                |
|                                                                                     | <i>RoleIDCode</i>                     | <i>U</i>  | Access Control role(s) the user holds that allows this transaction.                   |
|                                                                                     | <i>NetworkAccessPointTypeCode</i>     | <i>NA</i> |                                                                                       |
|                                                                                     | <i>NetworkAccessPointID</i>           | <i>NA</i> |                                                                                       |
| <b>Destination:<br/>AuditMessage/<br/>ActiveParticipant</b>                         | <i>UserID</i>                         | <i>M</i>  | SOAP Endpoint URI                                                                     |
|                                                                                     | <i>AlternativeUserID</i>              | <i>M</i>  | The process ID as used within the local operating system in the local system of logs  |
|                                                                                     | <i>UserName</i>                       | <i>U</i>  | <i>not specialized</i>                                                                |
|                                                                                     | <i>UsersRequestor</i>                 | <i>U</i>  | <i>not specialized</i>                                                                |
|                                                                                     | <i>RoleIDCode</i>                     | <i>M</i>  | EV (110152, DCM, "Destination Role ID")                                               |
|                                                                                     | <i>NetworkAccessPointTypeCode</i>     | <i>U</i>  | "1" for machine (DNS) name,<br>"2" for IP address                                     |
|                                                                                     | <i>NetworkAccessPointID</i>           | <i>U</i>  | The machine name or IP address                                                        |
| <b>Audit Source:<br/>AuditMessage/<br/>AuditSourceIdentification</b>                | <i>AlternativeUserID</i>              | <i>U</i>  | <i>not specialized</i>                                                                |
|                                                                                     | <i>UserName</i>                       | <i>U</i>  | <i>not specialized</i>                                                                |
|                                                                                     | <i>UsersRequestor</i>                 | <i>U</i>  | <i>not specialized</i>                                                                |
| <b>Requester Entity:<br/>AuditMessage/<br/>ParticipantObject<br/>Identification</b> | <i>ParticipantObjectTypeCode</i>      | <i>M</i>  | "1" (person)                                                                          |
|                                                                                     | <i>ParticipantObjectTypeCodeRole</i>  | <i>M</i>  | "11" (security user entity)                                                           |
|                                                                                     | <i>ParticipantObjectDataLifeCycle</i> | <i>U</i>  | <i>not specialized</i>                                                                |
|                                                                                     | <i>ParticipantObjectIDTypeCode</i>    | <i>M</i>  | Subject Role (value of the attribute "urn:oasis:names:tc:xacml:2.0:subject:role")     |
|                                                                                     | <i>ParticipantObjectSensitivity</i>   | <i>U</i>  | <i>not specialized</i>                                                                |
|                                                                                     | <i>ParticipantObjectID</i>            | <i>M</i>  | Subject ID (value of the attribute "urn:oasis:names:tc:xacml:1.0:subject:subject-id") |
|                                                                                     | <i>ParticipantObjectName</i>          | <i>U</i>  | <i>not specialized</i>                                                                |
|                                                                                     | <i>ParticipantObjectQuery</i>         | <i>U</i>  | <i>not specialized</i>                                                                |
|                                                                                     | <i>ParticipantObjectDetail</i>        | <i>U</i>  | <i>not specialized</i>                                                                |
| <b>Resource:<br/>AuditMessage/</b>                                                  | <i>ParticipantObjectTypeCode</i>      | <i>M</i>  | "2" (system)                                                                          |
|                                                                                     | <i>ParticipantObjectTypeCodeRole</i>  | <i>M</i>  | For CH:ADR due to XDS, MHD or RMU:<br>"3" (report)                                    |

|                                                |                                       |          |                                                                                                                                                 |
|------------------------------------------------|---------------------------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ParticipantObject Identification (1..n)</b> |                                       |          | For CH:ADR due to CH:PPQ or CH:PPQm:<br>"13" (security resource)<br>For CH:ADR due to ATC:<br>"17" (data repository)                            |
|                                                | <i>ParticipantObjectDataLifeCycle</i> | <i>U</i> | <i>not specialized</i>                                                                                                                          |
|                                                | <i>ParticipantObjectIDTypeCode</i>    | <i>U</i> | <i>not specialized</i>                                                                                                                          |
|                                                | <i>ParticipantObjectSensitivity</i>   | <i>U</i> | <i>not specialized</i>                                                                                                                          |
|                                                | <i>ParticipantObjectID</i>            | <i>M</i> | Resource-ID (value of the attribute "urn:oasis:names:tc:xacml:1.0:resource:resource-id")                                                        |
|                                                | <i>ParticipantObjectName</i>          | <i>U</i> | <i>not specialized</i>                                                                                                                          |
|                                                | <i>ParticipantObjectQuery</i>         | <i>U</i> | <i>not specialized</i>                                                                                                                          |
|                                                | <i>ParticipantObjectDetail</i>        | <i>C</i> | Authorization result (when available):<br>type = fixed string "decision",<br>value = one of "Permit", "Deny", "NotApplicable", "Indeterminate". |

Table 17: CH:ADR Authorization Decision Provider Audit Message

## 3.2 Privacy Policy Feed [CH:PPQ-1]

### 3.2.1 Scope

This transaction is used by the CH:PPQ Policy Source to add, update, or delete privacy policies and policy sets stored in the CH:PPQ Policy Repository.

### 3.2.2 Referenced Standards

Privacy Policy Feed [CH:PPQ-1] messages shall be transmitted using synchronous Web Services, according to the requirements specified in IHE ITI TF-2<sup>14</sup>, Appendix V.

- a. IHE ITI TF-2<sup>14</sup>, Appendix V "Web Services for IHE transactions"  
<https://profiles.ihe.net/ITI/TF/Volume2/ch-V.html>
- b. Assertions and Protocols for the OASIS Security Assertion Markup Language (SAML) V2.0  
<https://docs.oasis-open.org/security/saml/v2.0/saml-core-2.0-os.pdf>
- c. The home page of the "OASIS eXtensible Access Control Markup Language" technical committee: [https://www.oasis-open.org/committees/tc\\_home.php?wg\\_abbrev=xacml](https://www.oasis-open.org/committees/tc_home.php?wg_abbrev=xacml)  
references all XACML related protocols and specifications for implementers of this profile.
  - I. OASIS Multiple Resource Profile of XACML v2.0  
[https://docs.oasis-open.org/xacml/2.0/access\\_control-xacml-2.0-mult-profile-spec-os.pdf](https://docs.oasis-open.org/xacml/2.0/access_control-xacml-2.0-mult-profile-spec-os.pdf)
  - II. OASIS eXtensible Access Control Markup Language (XACML) v2.0  
[https://docs.oasis-open.org/xacml/2.0/access\\_control-xacml-2.0-core-spec-os.pdf](https://docs.oasis-open.org/xacml/2.0/access_control-xacml-2.0-core-spec-os.pdf)  
Please be aware of the errata of the specification document as published on the XACML technical committee home page:
  - III. Errata: [http://www.oasis-open.org/committees/download.php/26986/access\\_control-xacml-2.0-core-spec-os-errata.doc](http://www.oasis-open.org/committees/download.php/26986/access_control-xacml-2.0-core-spec-os-errata.doc)
  - IV. OASIS SAML 2.0 profile of XACML v2.0
  - V. (Original: [http://docs.oasis-open.org/xacml/2.0/access\\_control-xacml-2.0-core-spec-os.pdf](http://docs.oasis-open.org/xacml/2.0/access_control-xacml-2.0-core-spec-os.pdf))
  - VI. Please be aware of the errata of the specification document as published on the XACML technical committee home page:

<sup>14</sup> IHE IT Infrastructure (ITI) Technical Framework, Volume 2, Revision 20.0, August 4, 2023.

Errata: [www.oasis-open.org/committees/download.php/24681/xacml-profile-saml2.0-v2-spec-wd-5-en.pdf](http://www.oasis-open.org/committees/download.php/24681/xacml-profile-saml2.0-v2-spec-wd-5-en.pdf)

### 3.2.3 XML Namespaces

In addition to XML namespaces defined in IHE ITI TF-2<sup>15</sup>, Appendix V.2.4, the following namespaces and namespace prefixes will be used:

| Prefix        | Namespace                                                        | Specification                                                                                                |
|---------------|------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| epr           | urn:e-health-suisse:2015:policy-administration                   | XSD document is available for download from the website of the Federal Office of Public Health <sup>16</sup> |
| hl7           | urn:hl7-org:v3                                                   | HL7v3 specification on HL7.org                                                                               |
| saml          | urn:oasis:names:tc:SAML:2.0:assertion                            | Assertions and Protocols for the OA-SIS Security Assertion Markup Language (SAML) V2.0                       |
| xacml         | urn:oasis:names:tc:xacml:2.0:policy:schema:os                    | OASIS eXtensible Access Control Markup Language (XACML) v2.0                                                 |
| xacml-context | urn:oasis:names:tc:xacml:2.0:context:schema:os                   | OASIS eXtensible Access Control Markup Language (XACML) v2.0                                                 |
| xacml-saml    | urn:oasis:names:tc:xacml:2.0:profile:saml2.0:v2:schema:assertion | OASIS SAML 2.0 profile of XACML v2.0                                                                         |
| xacml-samlp   | urn:oasis:names:tc:xacml:2.0:profile:saml2.0:v2:schema:protocol  | OASIS SAML 2.0 profile of XACML v2.0                                                                         |

Table 18: XML Namespaces

### 3.2.4

#### Interaction Diagram

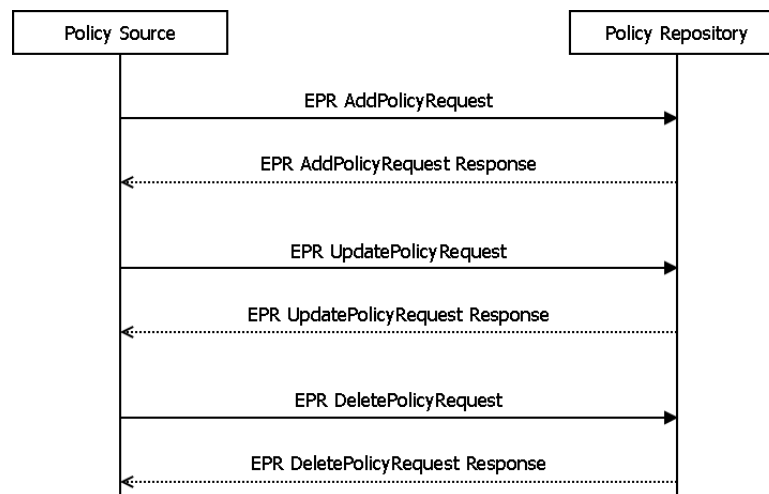


Figure 6: Sequence diagrams for the Privacy Policy Feed [CH:PPQ-1] transaction

<sup>15</sup> IHE IT Infrastructure (ITI) Technical Framework, Volume 2, Revision 20.0, August 4, 2023.

<sup>16</sup> Bundesamt für Gesundheit BAG, Gesetzgebung Elektronisches Patientendossier (EPDG), <https://www.bag.admin.ch/epdg-das-gesetz-hinter-dem-epd> > Schemata vom 1. Juni 2025 zur Ausgabe 7 der Ergänzung 2.1 zu Anhang 5 EPDV-EDI.zip

### 3.2.5 AddPolicyRequest and UpdatePolicyRequest

#### 3.2.5.1 Trigger Events

The CH:PPQ Policy Source sends these messages when it needs to add new or update existing policies and/or policy sets stored in the CH:PPQ Policy Repository.

#### 3.2.5.2 Message Semantics

The AddPolicyRequest and UpdatePolicyRequest request messages SHALL use SOAP 1.2 encoding and comply with all requirements described in IHE ITI TF-2<sup>17</sup>, Appendix V "Web Services for IHE Transaction".

The WS-Addressing header <wsa:Action> SHALL contain the value:

- a. For AddPolicyRequest: "urn:e-health-suisse:2015:policy-administration:AddPolicy".
- b. For UpdatePolicyRequest: "urn:e-health-suisse:2015:policy-administration:UpdatePolicy".

The WS-Security header <wse:Security> SHALL contain a CH:XUA Assertion.

The SOAP body SHALL contain a single element <epr:AddPolicyRequest> or <epr:UpdatePolicyRequest> (an EPR-specific XML schema epr-policy-administration-combined-schema-1.3-local.xsd is used, because the SAML 2.0 profile of XACML v2.0 does not define messages suitable for these requests). The only child of this root element SHALL be the element <saml:Assertion>.

The assertion in <saml:Assertion> SHALL contain following elements:

- c. Exactly one element <saml:Issuer> SHALL identify the CH:ADR Authorization Decision Provider by containing the home community ID of its community encoded as an URN and an XML attribute @NameQualifier equal to "urn:e-health-suisse:community-index".
- d. Exactly one element <saml:Statement> of the type "xacml-saml:XACMLPolicyStatementType" with one or more policies and policy sets contained in child elements <xacml:Policy> and <xacml:PolicySet>.

#### 3.2.5.3 Expected Actions

The CH:PPQ Policy Repository SHALL perform an Authorization Decision Request [CH:ADR] to determine whether the requesting user has the permission to fulfil the given CH:PPQ request.

If yes, the CH:PPQ Policy Repository SHALL add (for AddPolicyRequest) or update (for UpdatePolicyRequest) policies and policy sets contained in the CH:PPQ request, and create a CH:PPQ response message according to the success or failure of the transaction as defined below.

Otherwise, the CH:PPQ Policy Repository SHALL create a CH:PPQ error response message.

### 3.2.6 AddPolicyRequest Response and UpdatePolicyRequest Response

#### 3.2.6.1 Trigger Events

The AddPolicyRequest response or UpdatePolicyRequest response message is created by the CH:PPQ Policy Repository in response to the AddPolicyRequest or UpdatePolicyRequest message, respectively.

#### 3.2.6.2 Message Semantics

The AddPolicyRequest and UpdatePolicyRequest response messages SHALL use SOAP 1.2 encoding and comply with all requirements described in ITI TF-2<sup>17</sup>, chapter Appendix V "Web Services for IHE Transaction".

The WS-Addressing header <wsa:Action> SHALL contain the value:

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<sup>17</sup> IHE IT Infrastructure (ITI) Technical Framework, Volume 2, Revision 20.0, August 4, 2023.

- a. For AddPolicyRequest Response:  
"urn:e-health-suisse:2015:policy-administration:AddPolicyResponse".
- b. For UpdatePolicyRequest Response:  
"urn:e-health-suisse:2015:policy-administration:UpdatePolicyResponse".

The SOAP body SHALL contain a single element <epr:PolicyRepositoryResponse> with the attribute @status set to:

- c. On success: "urn:e-health-suisse:2015:response-status:success".
- d. On failure: "urn:e-health-suisse:2015:response-status:failure".

There is no notion of partial success in Privacy Policy Feed [CH:PPQ-1]. If at least one policy or policy set within the request cannot be added or updated, the entire request SHALL result in a failure response.

In case of an update failure due to unknown Policy Set IDs, a SOAP Fault SHALL be returned to the CH:PPQ Policy Source. The element <soap12:Detail> of this Fault SHALL contain a single element <epr:UnknownPolicySetId>.

### 3.2.7 DeletePolicyRequest

#### 3.2.7.1 Trigger Events

The CH:PPQ Policy Source sends this message when it needs to delete existing policies and policy sets from the CH:PPQ Policy Repository.

#### 3.2.7.2 Message Semantics

The DeletePolicyRequest request message SHALL use SOAP 1.2 encoding and comply with all requirements described in IHE ITI TF-2<sup>18</sup>, Appendix V "Web Services for IHE Transaction".

The WS-Addressing header <wsa:Action> SHALL contain the value:

urn:e-health-suisse:2015:policy-administration:DeletePolicy

The WS-Security header <wse:Security> SHALL contain a CH:XUA assertion.

The SOAP body SHALL contain a single element <epr:DeletePolicyRequest>. The only child of this root element SHALL be the element <saml:Assertion>.

The assertion in <saml:Assertion> SHALL contain following elements:

- a. Exactly one element <saml:Issuer> SHALL identify the CH:ADR Authorization Decision Provider by containing the home community ID of its community encoded as an URN and an XML attribute @NameQualifier equal to "urn:e-health-suisse:community-index".
- b. Exactly one element <saml:Statement> of the type "epr:XACMLPolicySetIdReferenceStatementType" with one or more policy IDs or policy set IDs contained in child elements <xacml:PolicySetIdReference>.

#### 3.2.7.3 Expected Actions

The CH:PPQ Policy Repository SHALL perform an Authorization Decision Request [CH:ADR] to determine whether the requesting user has the permission to fulfil the given CH:PPQ request.

If yes, the CH:PPQ Policy Repository SHALL delete policies and policy sets referenced in the CH:PPQ request, and create a CH:PPQ response message according to the success or failure of the transaction as defined below.

Otherwise, the CH:PPQ Policy Repository SHALL create a CH:PPQ error response message.

<sup>18</sup> IHE IT Infrastructure (ITI) Technical Framework, Volume 2, Revision 20.0, August 4, 2023.

### 3.2.8 DeletePolicyRequest Response

#### 3.2.8.1 Trigger Events

The DeletePolicyRequest response message is created by the CH:PPQ Policy Repository in response to the DeletePolicyRequest message.

#### 3.2.8.2 Message Semantics

The DeletePolicyRequest response message SHALL use SOAP 1.2 encoding and comply with all requirements described in ITI TF-2<sup>19</sup>, Appendix V "Web Services for IHE Transaction".

The WS-Addressing header <wsa:Action> SHALL contain the value:

`urn:e-health-suisse:2015:policy-administration:DeletePolicyResponse`

The SOAP body SHALL contain a single element <epr:PolicyRepositoryResponse> with the attribute @status set to:

- a. On success: "urn:e-health-suisse:2015:response-status:success".
- b. On failure: "urn:e-health-suisse:2015:response-status:failure".

There is no notion of partial success in Privacy Policy Feed [CH:PPQ-1]. If at least one policy or policy set referenced within the request cannot be deleted, the entire request SHALL result in a failure response.

In case of a deletion failure due to an unknown policy ID or policy set ID, a SOAP Fault SHALL be returned to the CH:PPQ Policy Source. The element <soap12:Detail> of this Fault SHALL contain a single element <epr:UnknownPolicySetId>.

### 3.2.9 Security Considerations

Relevant Security Considerations are defined in IHE ITI TF-1<sup>20</sup>, chapter 9.4. The CH:PPQ profile requires all actors to be grouped with Secure Node or Secure Application implementing the "STX: TLS 1.2 floor using BCP195 Option" defined in the IHE ITI TF-2<sup>21</sup>, chapter 3.19.6.2.3. Relevant XDS Affinity Domain Security background is discussed in the XDS Security Considerations section (see IHE ITI TF-1<sup>20</sup>, chapter 10.7). The involved actors SHALL record audit events according to the following definitions:

#### 3.2.9.1 CH:PPQ Policy Source Audit Message

|                                     | Field Name            | Opt | Value Constraints                                                                                     |
|-------------------------------------|-----------------------|-----|-------------------------------------------------------------------------------------------------------|
| <b>Event</b>                        | EventID               | M   | EV (110106, DCM, "Export")                                                                            |
|                                     | EventActionCode       | M   | a. For Add Policy: C = Create<br>b. For Update Policy: U = Update<br>c. For Delete Policy: D = Delete |
|                                     | EventDateTime         | M   | <i>not specialized</i>                                                                                |
|                                     | EventOutcomeIndicator | M   | <i>not specialized</i>                                                                                |
|                                     | EventTypeCode         | M   | EV("PPQ-1", "e-health-suisse", "Privacy Policy Feed")                                                 |
| Source (Policy Source) (1)          |                       |     |                                                                                                       |
| Human Requestor (0..n)              |                       |     |                                                                                                       |
| Destination (Policy Repository) (1) |                       |     |                                                                                                       |
| Audit Source (Policy Source) (1)    |                       |     |                                                                                                       |
| Patient (1..1)                      |                       |     |                                                                                                       |
| Policy or Policy Set (0..n)         |                       |     |                                                                                                       |

<sup>19</sup> IHE IT Infrastructure (ITI) Technical Framework, Volume 2, Revision 20.0, August 4, 2023.

<sup>20</sup> IHE IT Infrastructure (ITI) Technical Framework, Volume 1, Revision 20.0, August 4, 2023.

<sup>21</sup> IHE IT Infrastructure (ITI) Technical Framework, Volume 2, Revision 20.0, August 4, 2023.

|                                                                      |                                   |           |                                                                                      |
|----------------------------------------------------------------------|-----------------------------------|-----------|--------------------------------------------------------------------------------------|
| <b>Source:<br/>AuditMessage/<br/>ActiveParticipant</b>               | <i>UserID</i>                     | <i>U</i>  | <i>not specialized</i>                                                               |
|                                                                      | <i>AlternativeUserID</i>          | <i>M</i>  | The process ID as used within the local operating system in the local system of logs |
|                                                                      | <i>UserName</i>                   | <i>U</i>  | <i>not specialized</i>                                                               |
|                                                                      | <i>UsersRequestor</i>             | <i>U</i>  | <i>not specialized</i>                                                               |
|                                                                      | <i>RoleIDCode</i>                 | <i>M</i>  | EV (110153, DCM, "Source Role ID")                                                   |
|                                                                      | <i>NetworkAccessPointTypeCode</i> | <i>U</i>  | "1" for machine (DNS) name, "2" for IP address                                       |
|                                                                      | <i>NetworkAccessPointID</i>       | <i>U</i>  | The machine name or IP address.                                                      |
| <b>Human Requestor<br/>: AuditMessage/<br/>ActiveParticipant</b>     | <i>UserID</i>                     | <i>M</i>  | Identity of the human that initiated the transaction.                                |
|                                                                      | <i>AlternativeUserID</i>          | <i>U</i>  | <i>not specialized</i>                                                               |
|                                                                      | <i>UserName</i>                   | <i>U</i>  | <i>not specialized</i>                                                               |
|                                                                      | <i>UsersRequestor</i>             | <i>U</i>  | <i>not specialized</i>                                                               |
|                                                                      | <i>RoleIDCode</i>                 | <i>U</i>  | Access Control role(s) the user holds that allows this transaction.                  |
|                                                                      | <i>NetworkAccessPointTypeCode</i> | <i>NA</i> |                                                                                      |
|                                                                      | <i>NetworkAccessPointID</i>       | <i>NA</i> |                                                                                      |
| <b>Destination:<br/>AuditMessage/<br/>ActiveParticipant</b>          | <i>UserID</i>                     | <i>M</i>  | SOAP endpoint URI.                                                                   |
|                                                                      | <i>AlternativeUserID</i>          | <i>U</i>  | <i>not specialized</i>                                                               |
|                                                                      | <i>UserName</i>                   | <i>U</i>  | <i>not specialized</i>                                                               |
|                                                                      | <i>UsersRequestor</i>             | <i>U</i>  | <i>not specialized</i>                                                               |
|                                                                      | <i>RoleIDCode</i>                 | <i>U</i>  | EV (110152, DCM, "Destination Role ID")                                              |
|                                                                      | <i>NetworkAccessPointTypeCode</i> | <i>M</i>  | "1" for machine (DNS) name, "2" for IP address                                       |
|                                                                      | <i>NetworkAccessPointID</i>       | <i>M</i>  | The machine name or IP address.                                                      |
| <b>Audit Source:<br/>AuditMessage/<br/>AuditSourceIdentification</b> | <i>AlternativeUserID</i>          | <i>U</i>  | <i>not specialized</i>                                                               |
|                                                                      | <i>UserName</i>                   | <i>U</i>  | <i>not specialized</i>                                                               |
|                                                                      | <i>UsersRequestor</i>             | <i>U</i>  | <i>not specialized</i>                                                               |

|                                                                              |                                       |          |                                  |
|------------------------------------------------------------------------------|---------------------------------------|----------|----------------------------------|
| <b>Patient:<br/>(AuditMessage/<br/>ParticipantObject<br/>Identification)</b> | <i>ParticipantObjectTypeCode</i>      | <i>M</i> | "1" (person)                     |
|                                                                              | <i>ParticipantObjectTypeCodeRole</i>  | <i>M</i> | "1" (patient)                    |
|                                                                              | <i>ParticipantObjectDataLifeCycle</i> | <i>U</i> | <i>not specialized</i>           |
|                                                                              | <i>ParticipantObjectIDTypeCode</i>    | <i>M</i> | <i>not specialized</i>           |
|                                                                              | <i>ParticipantObjectSensitivity</i>   | <i>U</i> | <i>not specialized</i>           |
|                                                                              | <i>ParticipantObjectID</i>            | <i>M</i> | The patient ID in HL7 CX format. |
|                                                                              | <i>ParticipantObjectName</i>          | <i>U</i> | <i>not specialized</i>           |
|                                                                              | <i>ParticipantObjectQuery</i>         | <i>U</i> | <i>not specialized</i>           |
|                                                                              | <i>ParticipantObjectDetail</i>        | <i>U</i> | <i>not specialized</i>           |
| <b>Policy or Policy Set:<br/>AuditMessage/</b>                               | <i>ParticipantObjectTypeCode</i>      | <i>M</i> | "2" (system object)              |
|                                                                              | <i>ParticipantObjectTypeCodeRole</i>  | <i>M</i> | "13" (security resource)         |
|                                                                              | <i>ParticipantObjectDataLifeCycle</i> | <i>U</i> | <i>not specialized</i>           |
|                                                                              | <i>ParticipantObjectIDTypeCode</i>    | <i>M</i> | <i>not specialized</i>           |

|                                         |                                     |          |                                                                                                                                                                                                                    |
|-----------------------------------------|-------------------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ParticipantObject Identification</b> | <i>ParticipantObjectSensitivity</i> | <i>U</i> | <i>not specialized</i>                                                                                                                                                                                             |
|                                         | <i>ParticipantObjectID</i>          | <i>M</i> | ID of the policy or policy set from the CH:PPQ request message. For Add and Update requests, only IDs of top-level policies and policy sets are required; IDs of nested policies and policy sets should be omitted |
|                                         | <i>ParticipantObjectName</i>        | <i>U</i> | <i>not specialized</i>                                                                                                                                                                                             |
|                                         | <i>ParticipantObjectQuery</i>       | <i>U</i> | <i>not specialized</i>                                                                                                                                                                                             |
|                                         | <i>ParticipantObjectDetail</i>      | <i>U</i> | <i>not specialized</i>                                                                                                                                                                                             |

Table 19: CH:PPQ Policy Source Audit Message

## 3.2.9.2 CH:PPQ Policy Repository Audit Message

|                                                                                | Field Name                 | Opt | Value Constraints                                                                                     |
|--------------------------------------------------------------------------------|----------------------------|-----|-------------------------------------------------------------------------------------------------------|
| <b>Event</b>                                                                   | EventID                    | M   | EV (110107, DCM,"Import")                                                                             |
|                                                                                | EventActionCode            | M   | a. For Add Policy: C = Create<br>b. For Update Policy: U = Update<br>c. For Delete Policy: D = Delete |
|                                                                                | EventDateTime              | M   | <i>not specialized</i>                                                                                |
|                                                                                | EventOutcomeIndicator      | M   | <i>not specialized</i>                                                                                |
|                                                                                | EventTypeCode              | M   | EV("PPQ-1", "e-health-suisse", "Privacy Policy Feed")                                                 |
| Source (Policy Source) (1)                                                     |                            |     |                                                                                                       |
| Human Requestor (0..n)                                                         |                            |     |                                                                                                       |
| Destination (Policy Repository) (1)                                            |                            |     |                                                                                                       |
| Audit Source (Policy Repository) (1)                                           |                            |     |                                                                                                       |
| Patient (1..1)                                                                 |                            |     |                                                                                                       |
| Policy or Policy Set (0..n)                                                    |                            |     |                                                                                                       |
| <b>Source:<br/>AuditMessage/<br/>ActiveParticipant</b>                         | UserID                     | M   | <i>not specialized</i>                                                                                |
|                                                                                | AlternativeUserID          | U   | <i>not specialized</i>                                                                                |
|                                                                                | UserName                   | U   | <i>not specialized</i>                                                                                |
|                                                                                | UsersRequestor             | U   | <i>not specialized</i>                                                                                |
|                                                                                | RoleIDCode                 | M   | EV (110153, DCM, "Source Role ID")                                                                    |
|                                                                                | NetworkAccessPointTypeCode | M   | "1" for machine (DNS) name,<br>"2" for IP address                                                     |
|                                                                                | NetworkAccessPointID       | U   | The machine name or IP address.                                                                       |
| <b>Human Requestor<br/>(if known):<br/>AuditMessage/<br/>ActiveParticipant</b> | UserID                     | M   | Identity of the human that initiated the transaction.                                                 |
|                                                                                | AlternativeUserID          | U   | <i>not specialized</i>                                                                                |
|                                                                                | UserName                   | U   | <i>not specialized</i>                                                                                |
|                                                                                | UsersRequestor             | U   | <i>not specialized</i>                                                                                |
|                                                                                | RoleIDCode                 | U   | Access Control role(s) the user holds that allows this transaction.                                   |
|                                                                                | NetworkAccessPointTypeCode | NA  |                                                                                                       |
|                                                                                | NetworkAccessPointID       | NA  |                                                                                                       |
| <b>Destination:<br/>AuditMessage/<br/>ActiveParticipant</b>                    | UserID                     | M   | SOAP endpoint URI.                                                                                    |
|                                                                                | AlternativeUserID          | M   | The process ID as used within the local operating system in the local system of logs                  |

|                                                                                             |                                       |          |                                                                                                                                                                                                                                        |
|---------------------------------------------------------------------------------------------|---------------------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                             | <i>UserName</i>                       | <i>U</i> | <i>not specialized</i>                                                                                                                                                                                                                 |
|                                                                                             | <i>UserIsRequestor</i>                | <i>U</i> | <i>not specialized</i>                                                                                                                                                                                                                 |
|                                                                                             | <i>RoleIDCode</i>                     | <i>M</i> | EV (110152, DCM, "Destination Role ID")                                                                                                                                                                                                |
|                                                                                             | <i>NetworkAccessPointTypeCode</i>     | <i>U</i> | "1" for machine (DNS) name,<br>"2" for IP address                                                                                                                                                                                      |
|                                                                                             | <i>NetworkAccessPointID</i>           | <i>U</i> | The machine name or IP address.                                                                                                                                                                                                        |
| <b>Audit Source<br/>AuditMessage/<br/>AuditSourceIdentification</b>                         | <i>AlternativeUserID</i>              | <i>U</i> | <i>not specialized</i>                                                                                                                                                                                                                 |
|                                                                                             | <i>UserName</i>                       | <i>U</i> | <i>not specialized</i>                                                                                                                                                                                                                 |
|                                                                                             | <i>UserIsRequestor</i>                | <i>U</i> | <i>not specialized</i>                                                                                                                                                                                                                 |
| <b>Patient<br/>(AuditMessage/<br/>ParticipantObject<br/>Identification)</b>                 | <i>ParticipantObjectTypeCode</i>      | <i>M</i> | "1" (person)                                                                                                                                                                                                                           |
|                                                                                             | <i>ParticipantObjectTypeCodeRole</i>  | <i>M</i> | "1" (patient)                                                                                                                                                                                                                          |
|                                                                                             | <i>ParticipantObjectDataLifeCycle</i> | <i>U</i> | <i>not specialized</i>                                                                                                                                                                                                                 |
|                                                                                             | <i>ParticipantObjectIDTypeCode</i>    | <i>M</i> | <i>not specialized</i>                                                                                                                                                                                                                 |
|                                                                                             | <i>ParticipantObjectSensitivity</i>   | <i>U</i> | <i>not specialized</i>                                                                                                                                                                                                                 |
|                                                                                             | <i>ParticipantObjectID</i>            | <i>M</i> | The patient ID in HL7 CX format.                                                                                                                                                                                                       |
|                                                                                             | <i>ParticipantObjectName</i>          | <i>U</i> | <i>not specialized</i>                                                                                                                                                                                                                 |
|                                                                                             | <i>ParticipantObjectQuery</i>         | <i>U</i> | <i>not specialized</i>                                                                                                                                                                                                                 |
|                                                                                             | <i>ParticipantObjectDetail</i>        | <i>U</i> | <i>not specialized</i>                                                                                                                                                                                                                 |
| <b>Policy or Policy<br/>Set:<br/>AuditMessage/<br/>ParticipantObject<br/>Identification</b> | <i>ParticipantObjectTypeCode</i>      | <i>M</i> | "2" (system object)                                                                                                                                                                                                                    |
|                                                                                             | <i>ParticipantObjectTypeCodeRole</i>  | <i>M</i> | "13" (security resource)                                                                                                                                                                                                               |
|                                                                                             | <i>ParticipantObjectDataLifeCycle</i> | <i>U</i> | <i>not specialized</i>                                                                                                                                                                                                                 |
|                                                                                             | <i>ParticipantObjectIDTypeCode</i>    | <i>M</i> | <i>not specialized</i>                                                                                                                                                                                                                 |
|                                                                                             | <i>ParticipantObjectSensitivity</i>   | <i>U</i> | <i>not specialized</i>                                                                                                                                                                                                                 |
|                                                                                             | <i>ParticipantObjectID</i>            | <i>M</i> | ID of the policy or policy set from<br>the CH:PPQ request message. For<br>Add and Update requests, only IDs<br>of top-level policies and policy sets<br>are required; IDs of nested policies<br>and policy sets SHOULD be omit-<br>ted |
|                                                                                             | <i>ParticipantObjectName</i>          | <i>U</i> | <i>not specialized</i>                                                                                                                                                                                                                 |
|                                                                                             | <i>ParticipantObjectQuery</i>         | <i>U</i> | <i>not specialized</i>                                                                                                                                                                                                                 |
|                                                                                             | <i>ParticipantObjectDetail</i>        | <i>U</i> | <i>not specialized</i>                                                                                                                                                                                                                 |

Table 20: CH:PPQ Policy Repository Audit Message

### 3.3 Privacy Policy Retrieve [CH:PPQ-2]

#### 3.3.1 Scope

This transaction is used by the CH:PPQ Policy Consumer to retrieve privacy policies and policy sets from the CH:PPQ Policy Repository.

#### 3.3.2 Referenced Standards

Same as in the Privacy Policy Feed [CH:PPQ-1] transaction, see Section 3.2.2.

#### 3.3.3 XML Namespaces

Same as in the Privacy Policy Feed [CH:PPQ-1] transaction, see Section 3.2.3.

#### 3.3.4 Interaction Diagrams

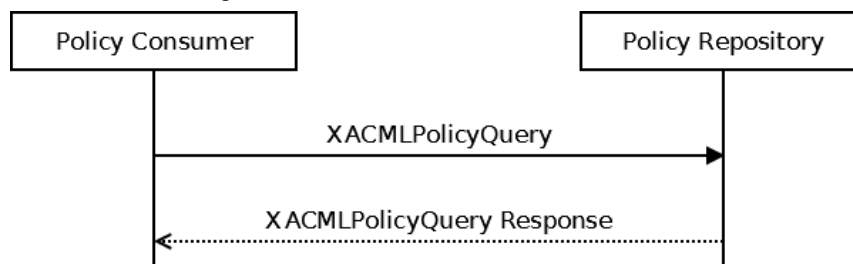


Figure 7: Sequence diagrams for the Privacy Policy Retrieve [CH:PPQ-2] transaction

#### 3.3.5 Policy Query Request

##### 3.3.5.1 Trigger Events

The CH:PPQ Policy Consumer sends this message when it needs to retrieve existing XACML policies or policy sets stored in a CH:PPQ Policy Repository.

##### 3.3.5.2 Message Semantics

The XACMLPolicyQuery request message SHALL use SOAP 1.2 encoding and comply with all requirements described in IHE ITI TF-2<sup>22</sup>, Appendix V "Web Services for IHE Transaction".

The WS-Addressing header <wsa:Action> SHALL contain the value:

```
urn:e-health-suisse:2015:policy-administration:PolicyQuery
```

The WS-Security header <wse:Security> SHALL contain a CH:XUA assertion.

The SOAP body SHALL contain a single element <xacml-samlp:XACMLPolicyQuery>.

Two different query flavors are supported:

- a. Retrieval of all policies and policy sets related to a particular patient. In this case, the only child of <xacml-samlp:XACMLPolicyQuery> SHALL be the element <xacml-context:Request>. Its elements <xacml-context:Subject>, <xacml-context:Action> and <xacml-context:Environment> SHOULD be empty as there is no use for them. Its element <xacml-context:Resource> SHALL contain a single child element <xacml-context:Attribute> with
  - I. XML attribute @AttributeId equal to "urn:e-health-suisse:2015:epr-spid",
  - II. XML attribute @DataType equal to of "urn:hl7-org:v3#II",
  - III. Contents of <xacml-context:AttributeValue> set to the EPR-SPID of the patient in HL7v3 Instance Identifier format, e.g.: <hl7:InstanceIdentifier xsi:type="hl7:II" root="2.16.756.5.30.1.127.3.10.3" extension="8901" />

<sup>22</sup> IHE IT Infrastructure (ITI) Technical Framework, Volume 2, Revision 20.0, August 4, 2023.

The request MAY contain more than one Resource, but they all SHALL reference the same patient.

- b. Retrieval of policies and policy sets directly referenced by their IDs (also useful for not patient-related policies). In this case, <xacml-samlp:XACMLPolicyQuery> SHALL contain one or more elements <xacml:PolicyIdReference> and/or <xacml:PolicySetIdReference> holding IDs of policies and policy sets, respectively.

### 3.3.5.3 Expected Actions

All policies satisfying the Resource definitions within a Request SHALL be returned if allowed by CH:ADR. Only policies and policy sets directly matching the search criteria shall be considered.

### 3.3.6 PolicyQuery Response

The PolicyQuery Response message is created by the CH:PPQ Policy Repository in response to the PolicyQuery Request.

#### 3.3.6.1 Trigger Events

This message is created by the CH:PPQ Policy Repository after the evaluation of a PolicyQuery Request message. The CH:PPQ Policy Repository identifies policies and policy sets applicable to be returned to the requester. Only policies and policy sets directly matching search criteria shall be considered, the ones referenced from them transitively SHALL NOT be returned.

#### 3.3.6.2 Message Semantics

The Policy Query Response message SHALL use SOAP 1.2 encoding and comply with all requirements described in IHE ITI TF-2<sup>23</sup>, Appendix V "Web Services for IHE Transaction".

The WS-Addressing header <wsa:Action> SHALL contain the value:

`urn:e-health-suisse:2015:policy-administration:PolicyQueryResponse`

The body of the SOAP message SHALL contain a single element <samlp:Response>, which SHALL contain a single element <samlp:Status> and a single element <saml:Assertion>:

The element <samlp:Status> contains a single element <samlp:StatusCode>, whose attribute @Value SHALL be set as defined in section 4.10 "Element <samlp:Response>: XACMLAuthzDecision Response" of OASIS SAML 2.0 profile of XACML v2.0 (errata).

The assertion in <saml:Assertion> SHALL contain following elements:

- a. Exactly one element <saml:Issuer> SHALL identify the CH:ADR Authorization Decision Provider by containing the home community ID of its community encoded as an URN and an XML attribute @NameQualifier equal to "urn:e-health-suisse:community-index".
- b. Exactly one element <saml:Statement> of the type "xacml-saml:XACMLPolicyStatement-Type" with one or more policies and policy sets contained in child elements <xacml:Policy> and <xacml:PolicySet>.

### 3.3.7 Security Considerations

Relevant Security Considerations are defined in IHE ITI TF-1<sup>24</sup>, chapter 9.4. The CH:PPQ profile requires all actors to be grouped with Secure Node or Secure Application implementing the "STX: TLS 1.2 floor using BCP195 Option" defined in the IHE ITI TF-2<sup>25</sup>, chapter 3.19.6.2.3. Relevant XDS Affinity Domain Security background is discussed in the XDS Security Considerations section (see IHE ITI TF-1<sup>24</sup>, chapter 10.7). The involved actors SHALL record audit events according to the following definitions:

<sup>23</sup> IHE IT Infrastructure (ITI) Technical Framework, Volume 2, Revision 20.0, August 4, 2023.

<sup>24</sup> IHE IT Infrastructure (ITI) Technical Framework, Volume 1, Revision 20.0, August 4, 2023.

<sup>25</sup> IHE IT Infrastructure (ITI) Technical Framework, Volume 2, Revision 20.0, August 4, 2023.

## 3.3.7.1 CH:PPQ Policy Consumer Audit Message

|                                                                      | Field Name                     | Opt | Value Constraints                                                                    |
|----------------------------------------------------------------------|--------------------------------|-----|--------------------------------------------------------------------------------------|
| <b>Event</b>                                                         | EventID                        | M   | EV (110112, DCM,"Query")                                                             |
|                                                                      | EventActionCode                | M   | E = Execute                                                                          |
|                                                                      | EventDateTime                  | M   | <i>not specialized</i>                                                               |
|                                                                      | EventOutcomeIndicator          | M   | <i>not specialized</i>                                                               |
|                                                                      | EventTypeCode                  | M   | EV("PPQ-2", "e-health-suisse", "Privacy Policy Retrieve")                            |
| Source (Policy Consumer) (1)                                         |                                |     |                                                                                      |
| Human Requestor (0..n)                                               |                                |     |                                                                                      |
| Destination (Policy Repository) (1)                                  |                                |     |                                                                                      |
| Audit Source (Policy Consumer) (1)                                   |                                |     |                                                                                      |
| Patient (1..1)                                                       |                                |     |                                                                                      |
| Query Parameters (1..1)                                              |                                |     |                                                                                      |
| <b>Source:<br/>AuditMessage/<br/>ActiveParticipant</b>               | UserID                         | U   | <i>not specialized</i>                                                               |
|                                                                      | AlternativeUserID              | M   | The process ID as used within the local operating system in the local system of logs |
|                                                                      | UserName                       | U   | <i>not specialized</i>                                                               |
|                                                                      | UsersRequestor                 | U   | <i>not specialized</i>                                                               |
|                                                                      | RoleIDCode                     | M   | EV (110153, DCM, "Source Role ID")                                                   |
|                                                                      | NetworkAccessPointTypeCode     | U   | "1" for machine (DNS) name, "2" for IP address                                       |
|                                                                      | NetworkAccessPointID           | U   | The machine name or IP address.                                                      |
| <b>Human Requestor:<br/>AuditMessage/<br/>ActiveParticipant</b>      | UserID                         | M   | Identity of the human that initiated the transaction.                                |
|                                                                      | AlternativeUserID              | U   | <i>not specialized</i>                                                               |
|                                                                      | UserName                       | U   | <i>not specialized</i>                                                               |
|                                                                      | UsersRequestor                 | U   | <i>not specialized</i>                                                               |
|                                                                      | RoleIDCode                     | U   | Access Control role(s) the user holds that allows this transaction.                  |
|                                                                      | NetworkAccessPointTypeCode     | NA  |                                                                                      |
|                                                                      | NetworkAccessPointID           | NA  |                                                                                      |
| <b>Destination:<br/>AuditMessage/<br/>ActiveParticipant</b>          | UserID                         | M   | SOAP endpoint URI.                                                                   |
|                                                                      | AlternativeUserID              | U   | <i>not specialized</i>                                                               |
|                                                                      | UserName                       | U   | <i>not specialized</i>                                                               |
|                                                                      | UsersRequestor                 | U   | <i>not specialized</i>                                                               |
|                                                                      | RoleIDCode                     | U   | EV (110152, DCM, "Destination Role ID")                                              |
|                                                                      | NetworkAccessPointTypeCode     | M   | "1" for machine (DNS) name, "2" for IP address                                       |
|                                                                      | NetworkAccessPointID           | M   | The machine name or IP address.                                                      |
| <b>Audit Source:<br/>AuditMessage/<br/>AuditSourceIdentification</b> | AlternativeUserID              | U   | <i>not specialized</i>                                                               |
|                                                                      | UserName                       | U   | <i>not specialized</i>                                                               |
|                                                                      | UsersRequestor                 | U   | <i>not specialized</i>                                                               |
| <b>Patient:<br/>(AuditMessage/</b>                                   | ParticipantObjectTypeCode      | M   | "1" (person)                                                                         |
|                                                                      | ParticipantObjectTypeCodeRole  | M   | "1" (patient)                                                                        |
|                                                                      | ParticipantObjectDataLifeCycle | U   | <i>not specialized</i>                                                               |

|                                                                                 |                                |   |                                                                                                                                                                 |
|---------------------------------------------------------------------------------|--------------------------------|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ParticipantObject Identification)</b>                                        | ParticipantObjectIDTypeCode    | M | <i>not specialized</i>                                                                                                                                          |
|                                                                                 | ParticipantObjectSensitivity   | U | <i>not specialized</i>                                                                                                                                          |
|                                                                                 | ParticipantObjectID            | M | The patient ID in HL7 CX format.                                                                                                                                |
|                                                                                 | ParticipantObjectName          | U | <i>not specialized</i>                                                                                                                                          |
|                                                                                 | ParticipantObjectQuery         | U | <i>not specialized</i>                                                                                                                                          |
|                                                                                 | ParticipantObjectDetail        | U | <i>not specialized</i>                                                                                                                                          |
| <b>Query Parameters:<br/>AuditMessage/<br/>ParticipantObject Identification</b> | ParticipantObjectTypeCode      | M | "2" (system object)                                                                                                                                             |
|                                                                                 | ParticipantObjectTypeCodeRole  | M | "24" (query)                                                                                                                                                    |
|                                                                                 | ParticipantObjectDataLifeCycle | U | <i>not specialized</i>                                                                                                                                          |
|                                                                                 | ParticipantObjectIDTypeCode    | M | <i>not specialized</i>                                                                                                                                          |
|                                                                                 | ParticipantObjectSensitivity   | U | <i>not specialized</i>                                                                                                                                          |
|                                                                                 | ParticipantObjectID            | M | Value of the attribute /XACMLPolicyQuery/@ID                                                                                                                    |
|                                                                                 | ParticipantObjectName          | U | <i>not specialized</i>                                                                                                                                          |
|                                                                                 | ParticipantObjectQuery         | M | The XACML Policy Query, base64 encoded                                                                                                                          |
|                                                                                 | ParticipantObjectDetail        | M | Attribute type — fixed string "QueryEncoding", attribute value — name the character encoding, such as "UTF-8", used to encode the query before base64 encoding. |

Table 21: CH:PPQ Policy Consumer Audit Message

## 3.3.7.2 CH:PPQ Policy Repository Audit Message

|                                                        | Field Name                 | Opt | Value Constraints                                         |
|--------------------------------------------------------|----------------------------|-----|-----------------------------------------------------------|
| <b>Event</b>                                           | EventID                    | M   | EV (110112, DCM, "Query")                                 |
|                                                        | EventActionCode            | M   | E = Execute                                               |
|                                                        | EventDateTime              | M   | <i>not specialized</i>                                    |
|                                                        | EventOutcomeIndicator      | M   | <i>not specialized</i>                                    |
|                                                        | EventTypeCode              | M   | EV("PPQ-2", "e-health-suisse", "Privacy Policy Retrieve") |
| Source (Policy Consumer) (1)                           |                            |     |                                                           |
| Human Requestor (0..n)                                 |                            |     |                                                           |
| Destination (Policy Repository) (1)                    |                            |     |                                                           |
| Audit Source (Policy Repository) (1)                   |                            |     |                                                           |
| Patient (1..1)                                         |                            |     |                                                           |
| Query Parameters (1..1)                                |                            |     |                                                           |
| <b>Source:<br/>AuditMessage/<br/>ActiveParticipant</b> | UserID                     | M   | <i>not specialized</i>                                    |
|                                                        | AlternativeUserID          | U   | <i>not specialized</i>                                    |
|                                                        | UserName                   | U   | <i>not specialized</i>                                    |
|                                                        | UsersRequestor             | U   | <i>not specialized</i>                                    |
|                                                        | RoleIDCode                 | M   | EV (110153, DCM, "Source Role ID")                        |
|                                                        | NetworkAccessPointTypeCode | M   | "1" for machine (DNS) name, "2" for IP address            |
|                                                        | NetworkAccessPointID       | U   | The machine name or IP address.                           |
| <b>Human Requestor:</b>                                | UserID                     | M   | Identity of the human that initiated the transaction.     |
|                                                        | AlternativeUserID          | U   | <i>not specialized</i>                                    |

|                                                                 |                                   |           |                                                                                      |
|-----------------------------------------------------------------|-----------------------------------|-----------|--------------------------------------------------------------------------------------|
| <b>AuditMessage/<br/>ActiveParticipant</b>                      | <i>UserName</i>                   | <i>U</i>  | <i>not specialized</i>                                                               |
|                                                                 | <i>UserIsRequestor</i>            | <i>U</i>  | <i>not specialized</i>                                                               |
|                                                                 | <i>RoleIDCode</i>                 | <i>U</i>  | Access Control role(s) the user holds that allows this transaction.                  |
|                                                                 | <i>NetworkAccessPointTypeCode</i> | <i>NA</i> |                                                                                      |
|                                                                 | <i>NetworkAccessPointID</i>       | <i>NA</i> |                                                                                      |
| <b>Destination:<br/>AuditMessage/<br/>ActiveParticipant</b>     | <i>UserID</i>                     | <i>M</i>  | SOAP endpoint URI.                                                                   |
|                                                                 | <i>AlternativeUserID</i>          | <i>M</i>  | The process ID as used within the local operating system in the local system of logs |
|                                                                 | <i>UserName</i>                   | <i>U</i>  | <i>not specialized</i>                                                               |
|                                                                 | <i>UserIsRequestor</i>            | <i>U</i>  | <i>not specialized</i>                                                               |
|                                                                 | <i>RoleIDCode</i>                 | <i>M</i>  | EV (110152, DCM, "Destination Role ID")                                              |
|                                                                 | <i>NetworkAccessPointTypeCode</i> | <i>U</i>  | "1" for machine (DNS) name, "2" for IP address                                       |
|                                                                 | <i>NetworkAccessPointID</i>       | <i>U</i>  | The machine name or IP address.                                                      |
| <b>Audit Source AuditMessage/<br/>AuditSourceIdentification</b> | <i>AlternativeUserID</i>          | <i>U</i>  | <i>not specialized</i>                                                               |
|                                                                 | <i>UserName</i>                   | <i>U</i>  | <i>not specialized</i>                                                               |
|                                                                 | <i>UserIsRequestor</i>            | <i>U</i>  | <i>not specialized</i>                                                               |

|                                                                                |                                       |          |                                                                                                                                                                               |
|--------------------------------------------------------------------------------|---------------------------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Patient<br/>(AuditMessage/<br/>ParticipantObjectIdentification)</b>         | <i>ParticipantObjectTypeCode</i>      | <i>M</i> | "1" (person)                                                                                                                                                                  |
|                                                                                | <i>ParticipantObjectTypeCodeRole</i>  | <i>M</i> | "1" (patient)                                                                                                                                                                 |
|                                                                                | <i>ParticipantObjectDataLifeCycle</i> | <i>U</i> | <i>not specialized</i>                                                                                                                                                        |
|                                                                                | <i>ParticipantObjectIDTypeCode</i>    | <i>M</i> | <i>not specialized</i>                                                                                                                                                        |
|                                                                                | <i>ParticipantObjectSensitivity</i>   | <i>U</i> | <i>not specialized</i>                                                                                                                                                        |
|                                                                                | <i>ParticipantObjectID</i>            | <i>M</i> | The patient ID in HL7 CX format.                                                                                                                                              |
|                                                                                | <i>ParticipantObjectName</i>          | <i>U</i> | <i>not specialized</i>                                                                                                                                                        |
|                                                                                | <i>ParticipantObjectQuery</i>         | <i>U</i> | <i>not specialized</i>                                                                                                                                                        |
|                                                                                | <i>ParticipantObjectDetail</i>        | <i>U</i> | <i>not specialized</i>                                                                                                                                                        |
| <b>Query Parameters:<br/>AuditMessage/<br/>ParticipantObjectIdentification</b> | <i>ParticipantObjectTypeCode</i>      | <i>M</i> | "2" (system object)                                                                                                                                                           |
|                                                                                | <i>ParticipantObjectTypeCodeRole</i>  | <i>M</i> | "24" (query)                                                                                                                                                                  |
|                                                                                | <i>ParticipantObjectDataLifeCycle</i> | <i>U</i> | <i>not specialized</i>                                                                                                                                                        |
|                                                                                | <i>ParticipantObjectIDTypeCode</i>    | <i>M</i> | <i>not specialized</i>                                                                                                                                                        |
|                                                                                | <i>ParticipantObjectSensitivity</i>   | <i>U</i> | <i>not specialized</i>                                                                                                                                                        |
|                                                                                | <i>ParticipantObjectID</i>            | <i>M</i> | Value of the attribute /XACMLPolicyQuery/@ID                                                                                                                                  |
|                                                                                | <i>ParticipantObjectName</i>          | <i>U</i> | <i>not specialized</i>                                                                                                                                                        |
|                                                                                | <i>ParticipantObjectQuery</i>         | <i>M</i> | The XACML Policy Query, base64 encoded                                                                                                                                        |
|                                                                                | <i>ParticipantObjectDetail</i>        | <i>M</i> | Attribute <i>type</i> — fixed string "QueryEncoding", attribute <i>value</i> — name the character encoding, such as "UTF-8", used to encode the query before base64 encoding. |

Table 22: CH:PPQ Policy Repository Audit Message

## 4 Volume 3 – Content Profiles - Submission Rules for Policies and Policy Sets

The collection of EPR base and template policies are available for download from the website of the Federal Office of Public Health<sup>26</sup>. It contains the following parts:

- a. Base Policies,
- b. Base Policy Sets,
- c. Templates for Patient Bootstrap Policy Sets,
- d. Templates for Patient User Assignment Policy Sets.

### 4.1 Base Policies and Base Policy Sets

Base Policies and Base Policy Sets have fixed IDs and are not related to a particular patient. They belong to the static configuration of a reference community and therefore SHALL be stored in the CH:PPQ Policy Repository.

### 4.2 Patient Bootstrap Policy Sets and Patient User Assignment Policy Sets

These two groups of policy sets relate to a particular patient.

The difference between those groups of policy sets is the following:

- a. **Bootstrap Policy Sets** SHALL be fed into the CH:PPQ Policy Repository during the on-boarding of a patient.
- b. **User Assignment Policy Sets** are created when
  - I. the patient grants permission to a healthcare professional, a group of healthcare professionals, or a representative,
  - II. the patient adds a healthcare professional onto the exclusion list,
  - III. a healthcare professional with delegation permission delegates access rights to another healthcare professional.

The CH:PPQ Policy Source SHALL create Patient Policies from the template policies by setting actual data (e.g. EPR-SPIDs of patients or references to Base Policy Sets) and generate policy set IDs as UUIDs.

Patient Policy Sets stored in and returned by a CH:PPQ Policy Repository SHALL be compliant to the templates and SHALL pass the Schematron validation available for download from the website of the Federal Office of Public Health<sup>26</sup>.

<sup>26</sup> Bundesamt für Gesundheit BAG, Gesetzgebung Elektronisches Patientendossier (EPDG), <https://www.bag.admin.ch/epdg-das-gesetz-hinter-dem-epd> > Schemata vom 1. Juni 2025 zur Ausgabe 7 der Ergänzung 2.1 zu Anhang 5 EPDV-EDI.zip

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